Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies[1].

☒ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper or thesis:

Disappearance of Spurious Valleys Via Overparameterization

Authored by:
If the work was compiled in a group, the names of all authors are required.

Last name(s):	**First name(s):**
PENC	MÁRK

With my signature I confirm the following:
- I have adhered to the rules set out in the Citation Guidelines.
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date	**Signature(s)**
ZÜRICH; CH; 2026.06.15.	

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

[1] For further information please consult the ETH Zurich websites, e.g. https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html and https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html (subject to change).

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Chair for Mathematical Information Science
Prof. Dr. H. Bölcskei

Student Project in Computer Science

Spring Semester 2026

Márk Penc

Disappearance of Spurious Valleys Via Overparameterization

Supervisors: Valentin Abadie
Prof. Dr. Helmut Bölcskei

June 2026

Abstract

Neural network loss landscapes are highly non-convex and can contain many non-global local minima. Despite this, local search methods such as gradient descent have empirically achieved remarkable success in finding low-loss solutions. A mathematical understanding of the potential barriers to optimization is therefore essential. In this work, we characterize the effect of overparameterization on spurious valleys, i.e., path connected components of a sub-level set that do not contain a global minimizer. We show that by doubling the width of a two layered neural network, every parameter becomes connected to a global minimizer via a non-increasing path. This doubling, we further show, is sharp. That is, with any smaller amount of overparameterization, spurious valleys may remain arbitrarily bad. Finally, for the squared L^2 loss, we show that under common initialization schemes, even without adding any neurons, the probability of reaching a global minimizer via a non-increasing path is at least one half.

Acknowledgments

I express my sincere gratitude to Valentin Abadie for his guidance and support throughout this project. His feedback, suggestions, and the many thought-provoking discussions we had have been invaluable to me. Under his supervision, I developed greater mathematical rigor and clarity, which will undoubtedly serve me well throughout my academic journey.

I am deeply grateful to Prof. Dr. Helmut Bölcskei for the opportunity to carry out this project at the Chair for Mathematical Information Science, and for his excellent courses, which have profoundly shaped my appreciation of the mathematical foundations underpinning this work.

I would also like to thank my friends and former colleagues at the Artificial Intelligence Laboratory at HUN-REN SZTAKI, who supported me in developing my interest in loss landscapes and overparameterization.

I am grateful to my friends at ETH Zürich, whose companionship made this semester even more rewarding, and with whom it was a pleasure to share my progress along the way.

Finally, I wish to thank my family, whose support has accompanied me throughout my studies abroad.

Contents

Abstract **i**

Acknowledgments **ii**

1 Introduction and Prior Work **1**
1.1 Introduction 1
1.2 Formal Preliminaries 2
1.2.1 Mathematical Spaces and $\mathbf{L}^p$ Spaces 2
1.2.2 Neural Networks 4
1.3 Prior Work 8
1.3.1 Properties of the Functional Equivalence Class 8
1.4 Contributions 10
1.5 Document Overview 12

Notation **14**

2 Assumptions and Properties of the Realization Set **16**
2.1 Assumptions 16
2.2 The Realization Set 18

3 Results on the Disappearance of Spurious Valleys **23**
3.1 Arbitrary Convex Loss in Function Space 23
3.2 $\mathbf{L}^2$ Distance in Function Space 27

4 Generalizations of Our Results **33**

5 Discussion **35**

A Proofs **39**

B Mathematical Preliminaries **47**

Chapter 1

Introduction and Prior Work

1.1 Introduction

Neural networks are parameterized functions that have gained widespread popularity due to their function approximation capabilities. Networks with sufficient capacity (width or depth) have been shown to approximate broad classes of functions with arbitrary precision [1, 2]. Moreover, they have achieved remarkable empirical success in computer vision [3], medical predictions [4], and mathematical discovery [5].

Despite this empirical success, many mathematical properties of neural networks remain poorly understood. Recent work has shown that neural networks can generalize well even when they have sufficient capacity to memorize the training data [6, 7]. This challenges the classical bias-variance tradeoff, which predicts that increasing model capacity beyond a certain point degrades generalization [8]. In recent years, this has led to a growing body of literature on overparameterization.

One such area of research studies how the loss landscape changes as the capacity of the network grows [9–12]. Classical work showed that for certain simple neural network architectures, minimal parameters, i.e., those realizing functions that cannot be realized with fewer neurons, are unique up to sign flips and neuron permutations [9, 13]. More recent work has extended these results by showing that non-minimal parameters, i.e., those whose function admits a realization with fewer neurons, form a connected affine variety [10, 11]. This line of research, known as neural network identifiability, has recently been resolved for a family of activation functions [12].

Another line of work studies the loss landscape from the perspective of spurious valleys, i.e., path connected components of a sub-level set that do not contain a global minimizer. Venturi et al. [14] showed that, under certain conditions, sufficiently wide networks with polynomial activation functions admit no spurious valleys. In contrast, networks with non-polynomial activation functions can exhibit arbitrarily bad spurious valleys.

Beyond theoretical analysis of the loss landscape, a parallel line of work investigates progressive network growth during training as an optimization technique. Methods that add or split neurons mid-training have been explored in both classical [15, 16] and modern [17, 18] settings, often with the explicit aim of escaping local minima.

In this work, we study whether local minima can be escaped via overparameter-

ization. Recall that extending a network with additional neurons yields a connected affine variety of functionally equivalent parameters. We investigate whether elements of this variety remain local minimizers, or whether sufficient overparameterization permits escape to a global minimizer, thereby eliminating spurious valleys. We prove that doubling the width suffices to escape every spurious valley. We further show that, in general, this doubling is sharp. With strictly less overparameterization, spurious valleys may remain arbitrarily bad. Finally, we prove that for the L^2 loss in function space, under common parameter initialization schemes, the probability of initializing in a spurious valley is at most one half.

In the rest of this chapter, we formalize the problem statement and introduce the necessary mathematical foundations of this work.

1.2 Formal Preliminaries

In this Section, we recall the formal preliminaries for this work. We start by stating the definition of L^p spaces. Thereafter, we introduce neural networks and describe some basic properties and architectures of them.

1.2.1 Mathematical Spaces and L^p Spaces

The following definitions are the ones used directly in the main text and assume a basic familiarity with measure theory. In Appendix B we go over a self-contained treatment of the topic from first principles, which subsumes this section. It introduces mathematical spaces, measure theory and the Lebesgue integral.

Definition 1.1. A Banach space is a complete normed space.

Definition 1.2. A Hilbert space is a complete inner product space.

Remark 1.1. Every inner product $\langle \cdot, \cdot \rangle$ on a vector space X induces a norm via $\|x\| \coloneqq \sqrt{\langle x, x \rangle}$. Hence, every Hilbert space is in particular a Banach space.

Remark 1.2. Let X be a normed space equipped with two norms $\|\cdot\|, \|\cdot\|'$. The norms are said to be equivalent if there exists $C_1, C_2 > 0$ such that for all $x \in X$ it holds that

$$C_1\|x\|' \leq \|x\| \leq C_2\|x\|'.$$

It can be shown that on $\mathbb{R}^m$ all norms are equivalent [19, Theorem 2.4-5]. In the following, we denote by $\|\cdot\|_{\mathbb{R}^m}$ an arbitrary norm on $\mathbb{R}^m$.

We now turn to L^p spaces, which provide the function-space setting for our analysis.

Definition 1.3. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. A property P is said to hold almost everywhere (a.e.) with respect to μ if there exists a set $S \in \mathcal{A}$ with $\mu(S) = 0$ such that P holds for all $\omega \in \Omega \setminus S$.

Definition 1.4. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. For $p \in \mathbb{N}_+$ the space L^p is defined as

$$L^p(\Omega, \mathcal{A}, \mu) \coloneqq \left\{ f : \Omega \to \mathbb{R} : f \text{ is measurable and } \int_\Omega |f|^p \, d\mu < \infty \right\},$$

where two elements f, g are considered equivalent if $f = g$ almost everywhere.

Definition 1.5. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $f : \Omega \to \mathbb{R}$. The L^p norm is defined by

$$\|f\|_p = \left(\int_\Omega |f|^p \, d\mu \right)^{\frac{1}{p}}.$$

Remark 1.3. It can be shown that the pair $(L^p, \|\cdot\|_p)$ forms a Banach space. Furthermore, if we define the inner product $\langle \cdot, \cdot \rangle_{L^2}$ on $L^2(\Omega, \mathcal{A}, \mu)$ as

$$\langle f, g \rangle_{L^2} = \int_\Omega (f \cdot g) \, d\mu,$$

the pair $(L^2(\Omega, \mathcal{A}, \mu), \langle \cdot, \cdot \rangle_{L^2})$ forms a Hilbert space.

We can generalize L^p spaces to vector valued functions in the following way.

Definition 1.6. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. For $p \in \mathbb{N}_+$ the space L^p of $\mathbb{R}^m$ valued functions is defined as

$$L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m) \coloneqq \left\{ f : \Omega \to \mathbb{R}^m : f \text{ is measurable and } \int_\Omega \|f\|_{\mathbb{R}^m}^p \, d\mu < \infty \right\},$$

where two elements f, g are considered equivalent if $f = g$ almost everywhere. By the equivalence of norms on $\mathbb{R}^m$, the choice of $\|\cdot\|_{\mathbb{R}^m}$ does not affect the space $L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m)$.

Definition 1.7. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $f : \Omega \to \mathbb{R}^m$ measurable. For $p \in \mathbb{N}_+$, the L^p norm is defined by

$$\|f\|_p \coloneqq \left(\int_\Omega \|f(\omega)\|_{\mathbb{R}^m}^p \, d\mu(\omega) \right)^{1/p}.$$

Remark 1.4. It can be shown that the pair $(L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m), \|\cdot\|_p)$ forms a Banach space. Furthermore, if we equip $\mathbb{R}^m$ with the Euclidean inner product $\langle y, z \rangle_{\mathbb{R}^m} = \sum_{i=1}^m y_i z_i$ and define the inner product $\langle \cdot, \cdot \rangle_{L^2}$ on $L^2(\Omega, \mathcal{A}, \mu; \mathbb{R}^m)$ by

$$\langle f, g \rangle_{L^2} \coloneqq \int_\Omega \langle f(\omega), g(\omega) \rangle_{\mathbb{R}^m} \, d\mu(\omega),$$

then $(L^2(\Omega, \mathcal{A}, \mu; \mathbb{R}^m), \langle \cdot, \cdot \rangle_{L^2})$ forms a Hilbert space.

1.2.2 Neural Networks

For all $p \in \mathbb{N}$ we define the parameter space as

$$\bar{\Theta}_p \coloneqq \mathbb{R}^{m\times p} \times \mathbb{R}^{p\times n} \times \mathbb{R}^p \times \mathbb{R}^m$$

Definition 1.8. A two layered neural network with width $p \in \mathbb{N}_+$ and activation function $\sigma : \mathbb{R} \to \mathbb{R}$ is a function $\bar{\mathcal{N}}_p : [0,1]^n \times \bar{\Theta}_p \to \mathbb{R}^m$. For an input $x \in [0,1]^n$ and a parameter $\theta = (U, W, b, d) \in \bar{\Theta}_p$, it is defined by

$$\bar{\mathcal{N}}_p(x;\theta) \coloneqq U\vec{\sigma}(Wx + b) + d.$$

The function $\vec{\sigma} : \mathbb{R}^p \to \mathbb{R}^p$ acts element wise on $z \in \mathbb{R}^p$ as

$$\vec{\sigma}(z) = (\sigma(z_1), \sigma(z_2), \ldots, \sigma(z_p))^\top.$$

Multiple classical results show that neural networks are universal function approximators. One such result is the following.

Theorem 1.1 (Universal Function Approximation [20], adapted to our setting)**.** Let $\sigma : \mathbb{R} \to \mathbb{R}$ be continuous and non-polynomial. Then for every continuous function $f : [0,1]^n \to \mathbb{R}^m$ and every $\epsilon > 0$, there exist $p \in \mathbb{N}_+$ and parameters $\theta \in \bar{\Theta}_p$ such that

$$\sup_{x\in[0,1]^n} \|\bar{\mathcal{N}}_p(x;\theta) - f(x)\|_2 < \epsilon.$$

Universal function approximation is a useful property, as it allows a single architecture to express functions to arbitrary precision.

For biasless neural networks, the parameter space is defined by

$$\Theta_p \coloneqq \mathbb{R}^{m\times p} \times \mathbb{R}^{p\times n}$$

Definition 1.9. A two layered biasless neural network with width $p \in \mathbb{N}_+$ and activation function $\sigma : \mathbb{R} \to \mathbb{R}$ is a function $\mathcal{N}_p : [0,1]^n \times \Theta_p \to \mathbb{R}^m$. For an input $x \in [0,1]^n$ and a parameter $\theta = (U, W) \in \Theta_p$, it is defined by

$$\mathcal{N}_p(x;\theta) \coloneqq \bar{\mathcal{N}}_p(x;(U, W, \mathbf{0}, \mathbf{0})) = U\vec{\sigma}(Wx).$$

Examples. Commonly used activation functions.

1. The linear function $\sigma(z) = z$.
2. The ReLU function defined by $\text{ReLU}(z) \coloneqq \max\{0, z\}$.
3. The hyperbolic tangent function

$$\sigma(z) = \frac{\exp(2z) - 1}{\exp(2z) + 1}$$

4. The sigmoid function

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

Similarly to [14], we define the set of functions that can be implemented by a neural network with a finite number of neurons. For simplicity, we define the following concepts for biasless neural networks. The concepts extend equivalently for neural networks with bias. In the following discussions, we will explicitly state when we deviate from our definitions and use the concepts for a biased neural network. Furthermore, throughout the rest of this section, unless otherwise stated, we assume that σ is continuous.

Definition 1.10. Let $p \in \mathbb{N}_+$, and consider an activation function $\sigma : \mathbb{R} \to \mathbb{R}$. The realization set of two layered, biasless neural networks with width p is defined as

$$V_{\sigma,p} \coloneqq \{\mathcal{N}_p(\cdot;\theta) : \theta \in \Theta_p\}$$

Definition 1.11. Consider an activation function $\sigma : \mathbb{R} \to \mathbb{R}$. The realization set of two layered, biasless neural networks is defined as

$$V_{\sigma,\infty} \coloneqq \bigcup_{p \in \mathbb{N}_+} V_{\sigma,p}$$

Definition 1.12. We denote by $\mathcal{U}$ the unit cube $[0,1]^n$ equipped with the Lebesgue measure

$$\mathcal{U} \coloneqq ([0,1]^n, B([0,1]^n), \lambda).$$

Proposition 1.1. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a continuous activation function. Then, for all $p \in \mathbb{N}_+$ we have

$$V_{\sigma,p} \subseteq L^2(\mathcal{U}; \mathbb{R}^m).$$

The proof can be found in Appendix A. Furthermore, this immediately implies

$$V_{\sigma,\infty} \subseteq L^2(\mathcal{U}; \mathbb{R}^m),$$

as $V_{\sigma,\infty}$ is defined as a union of subsets of the L^2 space.

Definition 1.13. Consider the measure space $\mathcal{U}$, let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a measurable function, and let $\sigma : \mathbb{R} \to \mathbb{R}$ be continuous. For all $p \in \mathbb{N}_+$, we define the functional equivalence class of f in Θ_p as

$$\mathcal{F}_{\mathcal{N}}(f,p) \coloneqq \{\theta \in \Theta_p : \mathcal{N}_p(\cdot;\theta) = f \ \lambda\text{-a.e. on } [0,1]^n\}$$

Definition 1.14. Consider the measure space $\mathcal{U}$. Let $q \in \mathbb{N}_+$, $\hat{\theta} \in \Theta_q$, and let $\sigma : \mathbb{R} \to \mathbb{R}$ be continuous. For all $p \in \mathbb{N}_+$, we define the functional equivalence class of $\hat{\theta}$ in Θ_p as

$$\mathcal{F}_{\mathcal{N}}(\hat{\theta}, p) \coloneqq \{\theta \in \Theta_p : \mathcal{N}_p(\cdot;\theta) = \mathcal{N}_q(\cdot;\hat{\theta}) \ \lambda\text{-a.e. on } [0,1]^n\}$$

Remark 1.5. In the above definition, due to the continuity of the activation function σ, our neural networks realize continuous maps. Therefore, $\mathcal{N}_p(\cdot;\theta) = \mathcal{N}_q(\cdot;\hat{\theta})$ λ-a.e. on $[0,1]^n$ if and only if for all $x \in [0,1]^n$ it holds that $\mathcal{N}_p(x;\theta) = \mathcal{N}_q(x;\hat{\theta})$.

Proof. Indeed, let $h = \mathcal{N}_p(\cdot;\theta) - \mathcal{N}_q(\cdot;\hat{\theta})$, and let $A = \{x \in [0,1]^n : h(x) \neq 0\}$. For the sake of contradiction, assume $A \neq \emptyset$, and let $x_0 \in A$. Since $h(x_0) \neq 0$ and h is continuous, there exists a relatively open neighborhood S of x_0 in $[0,1]^n$ with $S \subseteq A$. Since every nonempty relatively open subset of $[0,1]^n$ has positive Lebesgue measure, this contradicts $\mathcal{N}_p(\cdot;\theta) = \mathcal{N}_q(\cdot;\hat{\theta})$ λ-a.e. □

We say that a neural network parameter $\theta \in \Theta_p$ is minimal if $\mathcal{F}_{\mathcal{N}}(\theta, p) \neq \emptyset$ but $\mathcal{F}_{\mathcal{N}}(\theta, p-1) = \emptyset$. I.e., if it has the minimum number of neurons needed to realize the function.

In this work, we refer to non-minimal neural networks as overparameterized, and we refer to the overparameterization of a parameter as the embedding of it into a higher dimensional parameter space whilst implementing the same function.

Definition 1.15. The functional loss is a functional $l : L^2(\mathcal{U};\mathbb{R}^m) \to \mathbb{R}$.

Definition 1.16. The parameter loss is a function $\mathcal{L}_p : \Theta_p \to \mathbb{R}$ defined for $\theta \in \Theta_p$ by

$$\mathcal{L}_p(\theta) \coloneqq l(\mathcal{N}_p(\cdot;\theta))$$

Similarly to [14], we define the sub-level sets and spurious valleys as follows.

Definition 1.17. For $c \in \mathbb{R}$ and $p \in \mathbb{N}_+$ the sub-level set of $\mathcal{L}_p$ is defined as

$$S_c \coloneqq \{\theta \in \Theta_p : \mathcal{L}_p(\theta) \leq c\} \subseteq \Theta_p.$$

In order to discuss path connected components, we introduce the notion of path connectedness.

Definition 1.18. Let (X,d) be a metric space. A path in X is a continuous function $\gamma : [0,1] \to X$.

Lemma 1.2 (Pasting lemma)**.** Let (X,d) be a metric space and $\gamma_1, \gamma_2 : [0,1] \to X$ be two paths with $\gamma_1(1) = \gamma_2(0)$. Then $\gamma : [0,1] \to X$ defined as

$$\gamma(t) \coloneqq \begin{cases} \gamma_1(2t) & : t \in [0, 0.5] \\ \gamma_2(2t-1) & : t \in [0.5, 1] \end{cases}$$

is a path.

Proof. Indeed, for all $t \in [0, 0.5]$ and all $t \in [0.5, 1]$, we have $\gamma(t) \in X$. Furthermore, by the continuity of the paths γ_1, γ_2 we have that γ is continuous on $[0, 0.5)$ and $(0.5, 1]$. All that is left to show is that γ is continuous at $t = 0.5$. To this end, let $\epsilon > 0$ be arbitrary.

By the continuity of γ_1 at 1, there exists $\delta_1 > 0$ such that for every $s \in [0,1]$ with $|s-1| < \delta_1$,

$$d(\gamma_1(s), \gamma_1(1)) < \epsilon.$$

Likewise, by the continuity of γ_2 at 0, there exists $\delta_2 > 0$ such that for every $s \in [0,1]$ with $|s| < \delta_2$,

$$d(\gamma_2(s), \gamma_2(0)) < \epsilon.$$

Set $\delta \coloneqq \frac{1}{2}\min(\delta_1, \delta_2) > 0$ and let $t \in [0,1]$ with $|t - 0.5| < \delta$. If $t \in [0, 0.5]$, then $\gamma(t) = \gamma_1(2t)$, $\gamma(0.5) = \gamma_1(1)$, and

$$|2t - 1| = 2|t - 0.5| < 2\delta \leq \delta_1,$$

so $d(\gamma(t), \gamma(0.5)) < \epsilon$. The case $t \in [0.5, 1]$ is analogous, with γ_2 at 0 in place of γ_1 at 1. This establishes the continuity of γ at 0.5. □

Definition 1.19. Let (X, d) be a metric space, and $f : X \to \mathbb{R}$. A path $\gamma : [0,1] \to X$ is non-increasing with respect to f if for all $s, t \in [0,1]$ with $s \leq t$ it holds that

$$f(\gamma(t)) \leq f(\gamma(s)).$$

Definition 1.20. Let (X, d) be a metric space. Two points $x, y \in X$ are path connected in X if there exists a path $\gamma : [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Definition 1.21. Let (X, d) be a metric space. A path connected component of X is a maximal subset S of X for which any two points are path connected in S.

Now we can partition the sub-level set into path connected components.

Definition 1.22. Let $c \in \mathbb{R}$, $p \in \mathbb{N}_+$ and let $S_c \subseteq \Theta_p$ be the sub-level set of $\mathcal{L}_p$ with respect to c. We define the set of path connected components in S_c as

$$\mathcal{C}(S_c) \coloneqq \{A \subseteq S_c : A \text{ path connected component of } S_c\} \subseteq \mathcal{P}(\Theta_p)$$

Definition 1.23. Let $p \in \mathbb{N}_+$ and the global minimizer set be $\Theta^* \coloneqq \arg\min_\theta \mathcal{L}_p(\theta) \subseteq \Theta_p$. The set of spurious valleys, $\mathcal{V} \subseteq \mathcal{P}(\Theta_p)$ is then defined as

$$\mathcal{V} \coloneqq \bigcup_{c \in \mathbb{R}} \{C \in \mathcal{C}(S_c) : C \cap \Theta^* = \emptyset\}$$

We say that the loss admits no spurious valleys if $\mathcal{V} = \emptyset$.

In Figure 1.1 we show examples of different path connected components, some of which are spurious valleys, others are non-spurious. We show this for the same loss with two different sub level sets.

Remark 1.6. In [14] they introduced property **P.1**, which implies the absence of spurious valleys. Here we introduce a similar property localized to a parameter $\theta \in \Theta_p$.

P.1 A parameter $\theta \in \Theta_p$ satisfies **P.1** if there exists a path $\gamma : [0,1] \to \Theta_p$ satisfying the following.

1. $\gamma(0) = \theta$
2. $\gamma(1) \in \arg\min_{\theta \in \Theta_p} \mathcal{L}_p(\theta)$
3. γ is non-increasing with respect to $\mathcal{L}_p$

Lemma 1.3. Let $\mathcal{V} \subseteq \mathcal{P}(\Theta_p)$ be the set of spurious valleys with respect to the parameter loss $\mathcal{L}_p$. If a parameter $\theta \in \Theta_p$ satisfies Property **P.1** then for all $V \in \mathcal{V}$ we have $\theta \notin V$.

Proof. The proof can be found in Appendix A. □

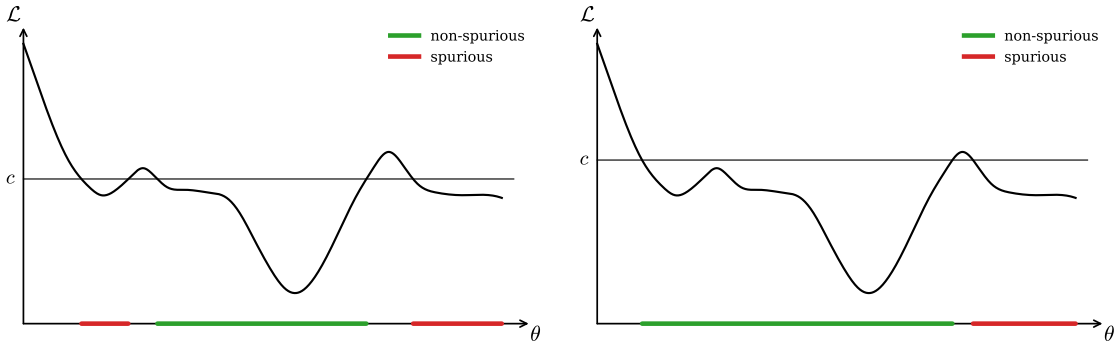


Figure 1.1: A loss surface with different sub-level sets. For a given c we color the sub-level set on the axis of the parameters. Red denotes spurious valleys while green denotes non-spurious valleys.

1.3 Prior Work

1.3.1 Properties of the Functional Equivalence Class

The question of neural network identifiability revolves around expressing the functional equivalence class $\mathcal{F}_{\mathcal{N}}(\theta, p)$.

The first paper characterizing the functional equivalence class of two layered biased neural networks of the form $\bar{\mathcal{N}}_p : \mathbb{R}^n \to \mathbb{R}$ with hyperbolic tangent activation function was [9]. In this setup, they showed that for a minimal neural network parameter $\theta \in \bar{\Theta}_p$ any other element of $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is obtained from θ by neuron permutations and/or sign-flips. The results were later extended in [13] to more general activation functions and Recurrent Neural Networks. Similar results have been derived for multilayer neural networks in [21, 12].

In [11] they investigated $\mathcal{F}_{\mathcal{N}}(\theta, p)$ for two layered neural networks $\bar{\mathcal{N}}_p : \mathbb{R}^n \times \bar{\Theta}_p \to \mathbb{R}^m$ with bias and with hyperbolic tangent activation function. Let $\theta \in \bar{\Theta}_q$ be a minimal neural network parameter. They showed that $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is path connected for $p > q$. Furthermore, the set $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is a union of affine subspaces. Similar results were achieved in [10] for a more general class of activation functions. They considered two layered neural networks of the form $\mathcal{N}_p : \mathbb{R}^n \times \Theta_p \to \mathbb{R}^m$ with σ continuous and $\sigma(x) \neq 0$ for all $x \in \mathbb{R}$. They showed that for a minimal neural network parameter $\theta \in \Theta_q$ it holds that $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is composed of a union of affine subspaces for $p > q$. Furthermore, the set is path-connected.

Example.

Consider for $p \in \{1, 2\}$ a two layered biasless neural network $\mathcal{N}_p : \mathbb{R} \times \Theta_p \to \mathbb{R}$ with sigmoid activation function. Let the minimal parameters be $\theta^* = (u^*, w^*) = (1, 2)$ so that

$$\mathcal{N}_1(x; \theta^*) = 1\sigma(2x).$$

In [10] they showed that $\mathcal{F}_{\mathcal{N}}(\theta^*, 2)$ consists of three affine subspaces K_1, K_2, K_3

defined as below.

$$K_1 = \left(\begin{pmatrix} u^* \\ 0 \end{pmatrix}, \begin{pmatrix} w^* \\ w \end{pmatrix} \right) : w \in \mathbb{R}$$
$$K_2 = \left(\begin{pmatrix} \alpha u^* \\ (1-\alpha)u^* \end{pmatrix}, \begin{pmatrix} w^* \\ w^* \end{pmatrix} \right) : \alpha \in \mathbb{R}$$
$$K_3 = \left(\begin{pmatrix} 0 \\ u^* \end{pmatrix}, \begin{pmatrix} w \\ w^* \end{pmatrix} \right) : w \in \mathbb{R}$$

Indeed, these are path connected as

$$K_1 \cap K_2 = \left\{ \left(\begin{pmatrix} u^* \\ 0 \end{pmatrix}, \begin{pmatrix} w^* \\ w^* \end{pmatrix} \right) \right\} =: \{B\} \neq \emptyset$$
$$K_2 \cap K_3 = \left\{ \left(\begin{pmatrix} 0 \\ u^* \end{pmatrix}, \begin{pmatrix} w^* \\ w^* \end{pmatrix} \right) \right\} =: \{C\} \neq \emptyset$$

Therefore, any two parameters in $\mathcal{F}_{\mathcal{N}}(\theta^*, 2)$ are connected via a path. Consider for the sake of example the parameters

$$A := \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right) \in K_1 \qquad D := \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right) \in K_3.$$

In Figure 1.2 we show a path between the two parameters.

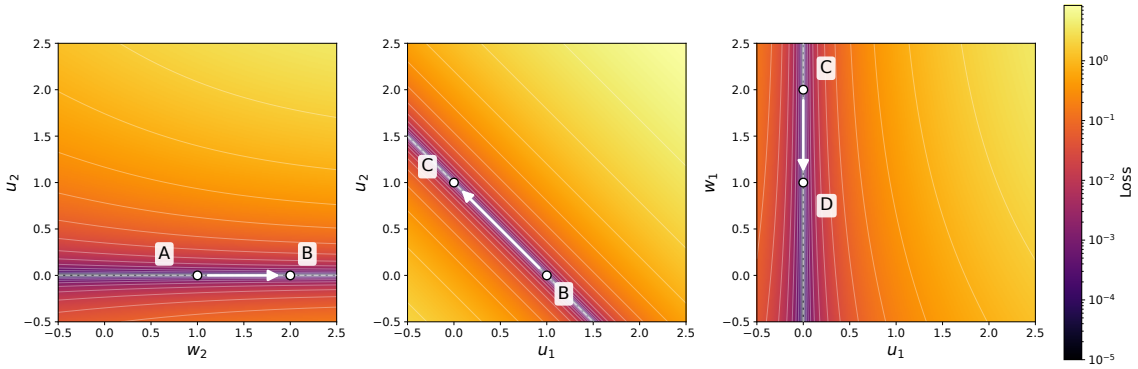


Figure 1.2: Two dimensional subspaces showing a heatmap of a loss surface $\mathcal{L}_2 : \mathbb{R}^4 \to \mathbb{R}$, with $\mathcal{L}_1(\theta^*) = 0$. We can see with zero loss the three subspaces mentioned in the example above, K_1, K_2, K_3, respectively. The subspaces intersect at the points B and C. There exists a path between the starting parameter A and the end parameter D as shown with white arrows.

In [14] they investigated conditions for which there is an absence of spurious valleys in the loss surface. They considered μ a probability distribution on the input data. Furthermore, σ is such that $\sigma(\langle w, \cdot \rangle) \in L^2(\mathbb{R}^n, B(\mathbb{R}^n), \mu)$ for all $w \in \mathbb{R}^n$. They defined the realization sets $V_{\sigma,p}$ and $V_{\sigma,\infty}$ w.r.t. the equivalence on the above mentioned L^2

space. Under some assumptions, they showed that for σ polynomial or μ discrete, the dimension of the realization set $V_{\sigma,\infty}$ is finite, otherwise infinite. Furthermore, they proved that in case we have more neurons than the dimension of the realization set, from any parameter there exists a path to a global minimizer with a non-increasing loss along it. Hence, there are no spurious valleys.

On the other hand, for all widths p, under some assumptions on the loss and the activation function (non-negative, non-polynomial and $\sigma \in L^2(\mathbb{R}^n, B(\mathbb{R}^n), e^{-x^2}dx)$), they constructed an adversarial data distribution such that there are arbitrarily bad spurious valleys. That is, for any $M > 0$, they showed that there exists a data distribution and parameters from which any path, $\gamma : [0,1] \to \Theta_p$, ending at a global minimizer must achieve a loss value greater than M at some $t_0 \in [0,1]$. Furthermore, the local minimizer in the spurious valley can be made arbitrarily far from the global minimizer. In particular, this holds for a wide range of activation functions, such as the ReLU or the Sigmoid. In Figure 1.3 we show an instance of a bad spurious valley.

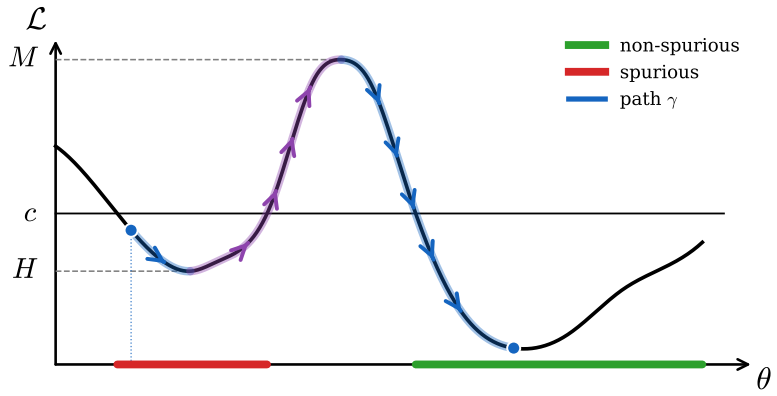


Figure 1.3: Spurious valleys can exist such that the local minimizer is arbitrarily far from the global one. Furthermore, any path to a global minima must pass through an arbitrarily high peak.

1.4 Contributions

Our problem can be formalized as follows. Let $\theta \in \Theta_q$ be a neural network parameter that is an element of a spurious valley. Consider extending the network with $k > 0$ additional neurons (e.g., by adding all zero rows and columns to (U, W)). From [11,10] we know that $\mathcal{F}_{\mathcal{N}}(\theta, q+k)$ will become a path connected component. We investigate geometrically whether $\mathcal{F}_{\mathcal{N}}(\theta, q+k)$ remains a spurious valley or whether, through the additional neurons, elements of $\mathcal{F}_{\mathcal{N}}(\theta, q+k)$ become path connected to elements of a global minimizer, thus eliminating spurious valleys.

Throughout our analysis of the problem, we will make certain assumptions to ease the proof structures and readability. However, our results can be extended to cover more general cases with ease. In Figure 1.4 we show an illustration of our problem statement.

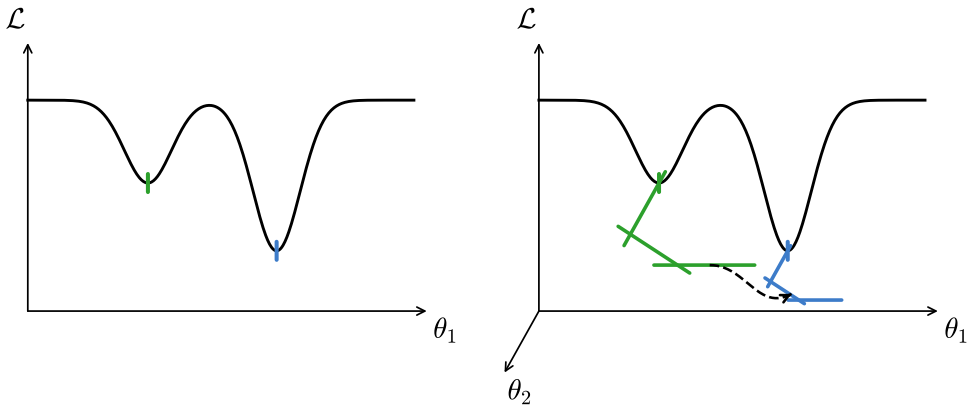


Figure 1.4: Consider a loss landscape seen on the left with one dimensional parameter space $\theta_1 \in \mathbb{R}$. It contains a spurious valley with a local minimum $\hat{\theta}$ as seen in green. The global minimizer θ^* is shown with blue. By extending the parameter space with additional parameters, θ_2, we get a functional equivalence class $\mathcal{F}_{\mathcal{N}}(\hat{\theta}, 2)$ (green) and $\mathcal{F}_{\mathcal{N}}(\theta^*, 2)$ (blue) on $\hat{\theta}$ and θ^* respectively. If from any point of $\mathcal{F}_{\mathcal{N}}(\hat{\theta}, 2)$ there exists a descent into any point of $\mathcal{F}_{\mathcal{N}}(\theta^*, 2)$ (dashed arrow), the spurious valley in lower dimension will not remain one in higher dimensions (conditioned on $\mathcal{F}_{\mathcal{N}}(\theta^*, 2)$ remaining a global minimizer).

For simplicity, one of our assumptions will be the realizability of the global minimizer. This is arguably the strongest assumption, but we emphasize that this can be relaxed. For the rest of this section, we will assume that

$$\arg\min_{\Phi \in V_{\sigma,\infty}} (l(\Phi)) \cap V_{\sigma,p} \neq \emptyset.$$

Definition 1.24. Let $p \in \mathbb{N}_+$. We say that a parameter $\theta \in \Theta_p$ is k-escapable if from every element of $\mathcal{F}_{\mathcal{N}}(\theta, p+k)$ there exists a non-increasing path with respect to $\mathcal{L}_{p+k}$ ending at an element of $\mathcal{F}_{\mathcal{N}}(\theta^*, p+k)$ for some $\theta^* \in \arg\min_{\theta \in \Theta_p} \mathcal{L}_p(\theta)$.

Question 1.1. Let μ be a probability measure on $(\Theta_p, B(\Theta_p))$ and $\theta \sim \mu$. What is the probability that θ is k-escapable? We denote this by

$$\varrho_p(k) := \mu(\{\theta \in \Theta_p \mid \theta \text{ is } k\text{-escapable}\}).$$

On measurability. A rigorous proof that the set above is measurable requires some descriptive set theory that we consider outside the scope of this work. The omission is benign as our main results are deterministic, and the one probabilistic result (Theorem 3.4) lower-bounds $\varrho_p(k)$ by the μ-measure of an explicit Borel subset of escapable parameters.

To the best of our knowledge, our problem statement is novel and has not been investigated in this setting yet. Our contributions can be summarized as follows.

1. **Doubling the width eliminates spurious valleys.** For any continuous convex functional loss and any width p, doubling the number of neurons suffices to escape every spurious valley. Under Assumptions A1–A4, from every parameter in $\mathcal{F}_{\mathcal{N}}(\theta, 2p)$ there is a path that is non-increasing with respect to $\mathcal{L}_{2p}$ and ends at a global minimizer (Theorem 3.1). Equivalently, every parameter is p-escapable, so $\varrho_p(p) = 1$, independently of the data distribution.

2. **Doubling the width is, in general, necessary.** We show this bound is sharp. For every width p and every $M > 0$ we construct parameters θ, θ^* and a convex functional loss for which adding only $p - 1$ neurons leaves an arbitrarily bad spurious valley. Every path from $\mathcal{F}_{\mathcal{N}}(\theta, 2p-1)$ to $\mathcal{F}_{\mathcal{N}}(\theta^*, 2p-1)$ attains a loss of at least M at some point (Theorem 3.2). Concentrating the initialization mass on these parameters yields $\varrho_p(p-1) = 0$, in contrast to $\varrho_p(p) = 1$.

3. **For the L^2 loss, at least half the initializations avoid spurious valleys.** Specializing to the squared L^2 loss $l(\Phi) = \|\Phi - f^*\|_{L^2}^2$, we show that under mild assumptions on the parameter-initialization distribution μ that common schemes satisfy, namely $U \overset{d}{=} -U$ and $U \perp\!\!\!\perp W$, at least half of all initializations are 0-escapable (Theorem 3.4). That is, $\varrho_p(0) \geq \frac{1}{2}$.

Importantly, we only show geometric properties of the loss landscape. We do not state anything with respect to optimization and gradient descent. Furthermore, our analysis only looks at the disappearance of spurious valleys with overparameterization, we do not investigate how new spurious valleys appear.

1.5 Document Overview

After going over common notations used in this work, we structure the document as follows.

- In Chapter 2 we go over properties of the realization set of two layered, biasless neural networks. We introduce multiple results that will prove to be useful for our later proofs.

- In Chapter 3 we detail the main results of this work relating to the disappearance of spurious valleys. We show that doubling the neurons of a parameter at a local minimum is enough to escape the functional equivalence class. We prove that this bound is a sharp one. Namely, there exist parameters and a convex loss where having strictly fewer than double the neurons does not eliminate spurious valleys. Furthermore, we investigate the case of the loss defined by the L^2 norm and derive that under common initialization schemes, even without additional neurons, we have greater than one-half probability of not initializing at a spurious valley.

- In Chapter 4 we go over how our results can be generalized further and how assumptions can be relaxed.

- In Chapter 5 we discuss our results and lay out potential future research directions.
- In Appendix A we detail some of the proofs that are longer or less relevant to the main text.
- In Appendix B we develop a self-contained introduction to mathematical spaces, the Lebesgue integral and L^p spaces.

Notation

By $\mathbb{N}_+$ we denote the set $\{1, 2, \ldots\}$ and by $\mathbb{R}_+$ the set $(0, \infty) \subseteq \mathbb{R}$. We denote the set $\{1, \ldots, n\}$ as $[n]$. For a metric space (S, d) a ball centered around $x \in S$ with radius ϵ is denoted by $\mathcal{B}(x, \epsilon) \coloneqq \{y \in S : d(x, y) \leq \epsilon\}$. We denote the Euclidean inner product on $\mathbb{R}^m$ by $\langle \cdot, \cdot \rangle_{\mathbb{R}^m}$ defined for $u, v \in \mathbb{R}^m$ by

$$\langle u, v \rangle_{\mathbb{R}^m} \coloneqq \sum_{i=1}^{m} u_i v_i.$$

Furthermore, we equip $\mathbb{R}^m$ with the norm $\| \cdot \|_2$ defined by

$$\|u\|_2 \coloneqq \left(\langle u, u \rangle_{\mathbb{R}^m} \right)^{\frac{1}{2}}.$$

We denote the zero vector in a vector space as $\mathbf{0}$.

For a metric space (X, d), we denote by $B(X)$ the Borel sigma field on X. For the measurable space $(\mathbb{R}^n, B(\mathbb{R}^n))$ we represent the Lebesgue measure by λ. By $L^2([0,1]^n; \mathbb{R}^m)$ we denote the L^2 space on the unit cube equipped with the Borel σ-field and Lebesgue measure,

$$L^2([0,1]^n, B([0,1]^n), \lambda; \mathbb{R}^m).$$

We equip the Hilbert space $L^2([0,1]^n; \mathbb{R}^m)$ with the inner product $\langle \cdot, \cdot \rangle_{L^2}$ defined for $f, g \in L^2([0,1]^n; \mathbb{R}^m)$ by

$$\langle f, g \rangle_{L^2} \coloneqq \int_{[0,1]^n} \langle f(x), g(x) \rangle_{\mathbb{R}^m} d\lambda(x).$$

We denote the norm induced by $\langle \cdot, \cdot \rangle_{L^2}$ by $\| \cdot \|_{L^2}$ and define it for $f \in L^2([0,1]^n; \mathbb{R}^m)$ as

$$\|f\|_{L^2} \coloneqq \left(\langle f, f \rangle_{L^2} \right)^{\frac{1}{2}}.$$

For two random variables X and Y, we write $X \overset{d}{=} Y$ to denote equality in distribution, that is, $\mathbb{P}_X = \mathbb{P}_Y$. For random variables X and Y with values in $(E_X, \mathcal{E}_X)$ and $(E_Y, \mathcal{E}_Y)$ respectively, we say that X and Y are independent, written $X \perp\!\!\!\perp Y$, if for all $A \in \mathcal{E}_X$ and $B \in \mathcal{E}_Y$ it holds that

$$\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A)\, \mathbb{P}(Y \in B).$$

Throughout the proofs we frequently group the neurons of a parameter and, with a slight abuse of notation, display the result as a single block array. For a parameter

$\theta = (U, W) \in \Theta_p$ whose p neurons are split into two groups, we write

$$\theta = \begin{pmatrix} U_1 & W_1 \\ U_2 & W_2 \end{pmatrix}$$

to denote the parameter with output-weight matrix $U = \begin{pmatrix} U_1 & U_2 \end{pmatrix}$ and input-weight matrix $W = \begin{pmatrix} W_1 \\ W_2 \end{pmatrix}$. More generally, for a partition of the neurons into k groups of sizes $p_1, \dots, p_k \in \mathbb{N}_+$ with $\sum_{j=1}^{k} p_j = p$, we write

$$\theta = \begin{pmatrix} U_1 & W_1 \\ U_2 & W_2 \\ \vdots & \vdots \\ U_k & W_k \end{pmatrix}, \qquad U_j \in \mathbb{R}^{m \times p_j}, \quad W_j \in \mathbb{R}^{p_j \times n},$$

to denote the parameter (U, W) obtained by the column concatenation $U = \begin{pmatrix} U_1 & \cdots & U_k \end{pmatrix} \in \mathbb{R}^{m \times p}$ and the row concatenation $W = \begin{pmatrix} W_1 \\ \vdots \\ W_k \end{pmatrix} \in \mathbb{R}^{p \times n}$. The j-th row $\begin{pmatrix} U_j & W_j \end{pmatrix}$ thus collects the output and input weights of the j-th block of neurons, and is itself identified with the sub-parameter $\theta_j = (U_j, W_j) \in \Theta_{p_j}$. We emphasize that this is purely notational, the blocks U_j and W_j have m and p_j rows respectively, so the displayed array is not a genuine matrix.

Chapter 2

Assumptions and Properties of the Realization Set

2.1 Assumptions

Throughout this work, we make some assumptions with respect to the activation function, the functional equivalence class, the global minimizer, and the realization set. Some of these are purely for the sake of simplification, while others are structurally necessary. Throughout the rest of the document, unless otherwise stated, we assume that the following hold.

Assumption A1. The activation function $\sigma : \mathbb{R} \to \mathbb{R}$ satisfies the following.

- σ is non-polynomial.
- σ is continuous.

The activation functions used for neural networks typically satisfy the above conditions. These include the hyperbolic tangent, ReLU, and variations of ReLU. Indeed, for polynomial activation functions, we have that universal function approximation does not hold [20], hence they are mostly used for analysis of algorithms. Furthermore, continuous functions are widely used because they produce continuous neural network outputs.

Assumption A2. The functional loss $l : L^2([0,1]^n; \mathbb{R}^m) \to \mathbb{R}$ is continuous and convex. That is, for all $\Phi_1, \Phi_2 \in L^2([0,1]^n; \mathbb{R}^m)$, and for all $t \in [0,1]$ it holds that

$$l((1-t)\Phi_1 + t\Phi_2) \leq (1-t)l(\Phi_1) + tl(\Phi_2)$$

Most loss functions commonly used are convex in the function space. Many of them are based on norms, such as the mean squared error $l(\Phi) = \|\Phi - f^*\|^2_{L^2(\mu)}$ for some measure μ on $([0,1]^n, B([0,1]^n))$ and $f^* \in L^2([0,1]^n; \mathbb{R}^m)$. We remark that convexity in function space does not imply convexity in parameter space.

Assumption A3. For a $\theta \in \Theta_q$ the functional equivalence class $\mathcal{F}_{\mathcal{N}}(\theta, p)$ is path-connected when $p > q$.

As discussed in the prior work section, this has been shown for certain activation functions. Furthermore, much of our analysis relies on moving first to a parameter that is equivalent to extending the network with all zero neurons. Hence, if the problem is phrased with respect to neuron extension, then this assumption is no longer necessary.

Assumption A4. When looking at if a $\theta \in \Theta_p$ is k-escapable, we have that the global minimizer across all widths is achieved for some parameters. In particular,

$$\arg\min_{\Phi \in V_{\sigma,\infty}} (l(\Phi)) \cap V_{\sigma,p} \neq \emptyset$$

This assumption is purely for the simplification of the problem. It is needed so that the minimizer to which we escape is already a global minimizer, and hence we can talk about the disappearance of spurious valleys. Due to Assumption A2 (convexity of functional loss) we will see in Chapter 4 that we can essentially escape even if Assumption A4 does not hold.

Assumption A5. For all $p \in \mathbb{N}_+$ when looking at if a $\theta \in \Theta_p$ is k-escapable, we have that

$$V_{\sigma,p} \not\subseteq \overline{V_{\sigma,p-1}}.$$

Intuitively, $V_{\sigma,p}$ contains a function that cannot be approximated to arbitrary precision by elements of $V_{\sigma,p-1}$. In [14] they showed that for neural networks mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$ with $n > 1$, $\sigma \in L^2(\mathbb{R}^n, B(\mathbb{R}^n), e^{-x^2/2}dx)$ continuous and non-polynomial this property holds when restricted to positive weights. We believe this property to hold with minimal assumptions in our setting as well, but proving it is beyond the scope of this work, hence we leave it as an assumption.

Although we do not prove this, we detail some ideas for how one could go about solving this.

1. In [14,22] they tackled this problem by looking at the symmetric tensor rank and arriving at a contradiction for polynomial activation functions with few neurons. Then they approximated the non-polynomial activations by Hermite polynomials to arrive at a contradiction in the case of non-polynomial activation functions. One could try to directly adopt their proof style to our setup.

2. Another possibility would be to use their results directly in our setup. That is, show due to their results, Assumption A5 holds. A tool for this would be to scale up the weights W in order to pull more of the function into the region $[0,1]^n$. Indeed, under their assumptions there is a region on $\mathbb{R}^n$ where the functions disagree. One could try to show that they disagree even in the range $[0,1]^n$. (might require moving to $[-1,1]^n$ as the input)

3. Another option would be to work with biased neural networks and try deriving a contradiction with universal function approximation. One could work towards showing that

   $$\overline{V_{\sigma,p-1}} = V_{\sigma,p} \quad \Rightarrow \quad \overline{V_{\sigma,p}} = V_{\sigma,p+1}$$

   and that this would imply that for some finite p we have $V_{\sigma,\infty} \subseteq \overline{V_{\sigma,p}}$ and hence, can approximate functions universally leading to a contradiction with established literature.

Finally, if none of these proofs work, one could try to adapt the entire setting from [14], since our results can be transferred to their setting.

Remark 2.1. Under Assumption A5, for all $p \in \mathbb{N}_+$ there exists a $\Phi \in V_{\sigma,p}$ and an $\epsilon > 0$ with

$$\mathcal{B}(\Phi, \epsilon) \cap \overline{V_{\sigma,p-1}} = \emptyset.$$

Proof. By Assumption A5 we have $V_{\sigma,p} \setminus \overline{V_{\sigma,p-1}} \neq \emptyset$, so we may fix some

$$\Phi \in V_{\sigma,p} \quad \text{with} \quad \Phi \notin \overline{V_{\sigma,p-1}}.$$

We claim some $\epsilon > 0$ has the desired property. Suppose, for contradiction, that no such ϵ exists. That is, for every $\epsilon > 0$,

$$\mathcal{B}(\Phi, \epsilon) \cap \overline{V_{\sigma,p-1}} \neq \emptyset.$$

Then for each $i \in \mathbb{N}_+$, applying this with $\epsilon = \frac{1}{i}$, we may pick

$$\Phi_i \in \mathcal{B}\big(\Phi, \tfrac{1}{i}\big) \cap \overline{V_{\sigma,p-1}}.$$

By construction $\Phi_i \in \overline{V_{\sigma,p-1}}$ and $\|\Phi_i - \Phi\|_{L^2} \leq \frac{1}{i}$, so the sequence $(\Phi_i)_{i \in \mathbb{N}_+}$ converges to Φ. Since $\overline{V_{\sigma,p-1}}$ is closed and each Φ_i lies in it, its limit lies in it as well, i.e. $\Phi \in \overline{V_{\sigma,p-1}}$. This contradicts the choice of Φ. Hence there exists $\epsilon > 0$ with $\mathcal{B}(\Phi, \epsilon) \cap \overline{V_{\sigma,p-1}} = \emptyset$. □

2.2 The Realization Set

First, we start by exploring certain properties of the realization set, with a particular focus on convexity. These results are crucial in the proofs later on.

Property 2.1. For all p we have $V_{\sigma,p} \subseteq V_{\sigma,p+1}$.

Proof. Indeed, any neural network that can be realized via p neurons can be realized by $p + 1$ by setting the weights of the last neuron to zero. In particular, let $\Phi \in V_{\sigma,p}$ be realized by $\theta = (U, W) \in \Theta_p$ then

$$\Phi(x) = U\vec{\sigma}(Wx) = \begin{pmatrix} U & \mathbf{0} \end{pmatrix} \vec{\sigma}\left(\begin{pmatrix} W \\ \mathbf{0}^\top \end{pmatrix} x \right) =: U'\vec{\sigma}(W'x)$$

Hence, $\Phi = \mathcal{N}_{p+1}(\cdot, (U', W')) \in V_{\sigma,p+1}$. □

Under Assumption A5, together with Property 2.1 the following is implied.

Corollary 2.1. Under Assumption A5, for all p we have $V_{\sigma,p} \subsetneq V_{\sigma,p+1}$.

The above is directly implied by 2.1 and Assumption A5.

Property 2.2. Consider $\theta_1 = (U_1, W) \in \Theta_p$ and $\theta_2 = (U_2, W) \in \Theta_p$. For the linear interpolation between θ_1 and θ_2 the following holds.

$$\mathcal{N}_p(\cdot; (1-t)\theta_1 + t\theta_2) = (1-t)\mathcal{N}_p(\cdot; \theta_1) + t\mathcal{N}_p(\cdot; \theta_2)$$

Proof. Indeed, for any $x \in [0,1]^n$ we have that

$$\begin{aligned}\mathcal{N}_p(x;(1-t)\theta_1 + t\theta_2) &= ((1-t)U_1 + tU_2)\vec{\sigma}(((1-t)W + tW)x) \\ &= ((1-t)U_1 + tU_2)\vec{\sigma}(Wx) \\ &= (1-t)U_1\vec{\sigma}(Wx) + tU_2\vec{\sigma}(Wx) \\ &= (1-t)\mathcal{N}_p(x;\theta_1) + t\mathcal{N}_p(x;\theta_2)\end{aligned}$$

□

This result implies that if the functional loss l is convex, then in the parameter space we have directions along which the loss function is convex. We illustrate this in Figure 2.1.

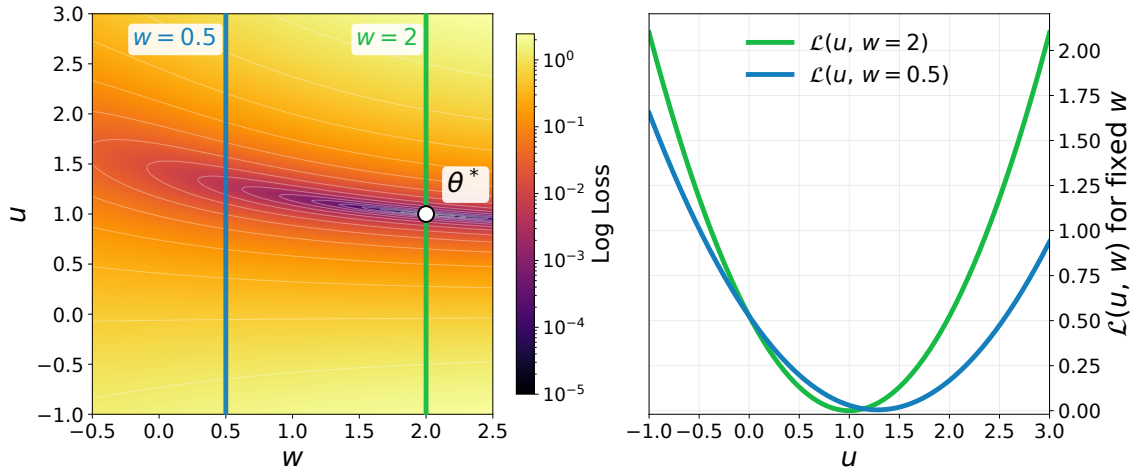


Figure 2.1: We plot the loss surface of a width one neural network. Importantly, keeping w fixed yields a convex loss when varying u.

With our current results, we can show that the set $V_{\sigma,p}$ is non-convex.

Property 2.3. Under Assumption A5 for all width p, the set $V_{\sigma,p}$ is non-convex.

Proof. As per Corollary 2.1, take $\Phi \in V_{\sigma,2p} \setminus V_{\sigma,2p-1} \neq \emptyset$. Since $\Phi \in V_{\sigma,2p}$ we can represent it with $2p$ neurons, which we denote by $\theta \in \Theta_{2p}$, i.e., $\mathcal{N}_{2p}(\cdot;\theta) = \Phi$.

Denote the first p neurons of θ by $\theta_1 = (U_1, W_1)$ and the last p by $\theta_2 = (U_2, W_2)$. In this way, we have that

$$\theta = \begin{pmatrix} U & W \end{pmatrix} = \begin{pmatrix} U_1 & W_1 \\ U_2 & W_2 \end{pmatrix} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}.$$

Now consider the parameters

$$\hat{\theta}_1 = \begin{pmatrix} 2U_1 & W_1 \\ 0 & W_2 \end{pmatrix} \quad \text{and} \quad \hat{\theta}_2 = \begin{pmatrix} 0 & W_1 \\ 2U_2 & W_2 \end{pmatrix}.$$

Notice that for $i \in \{1,2\}$ we have $\Phi_i := \mathcal{N}_{2p}(\cdot;\hat{\theta}_i) \in V_{\sigma,p}$, as for all x the following holds.

$$\Phi_1(x) = \begin{pmatrix} 2U_1 & 0 \end{pmatrix} \vec{\sigma}\left(\begin{pmatrix} W_1 \\ W_2 \end{pmatrix} x \right) = 2U_1 \, \vec{\sigma}(W_1 x) = \mathcal{N}_p\big(x; (2U_1, W_1)\big) \in V_{\sigma,p}$$
$$\Phi_2(x) = \begin{pmatrix} 0 & 2U_2 \end{pmatrix} \vec{\sigma}\left(\begin{pmatrix} W_1 \\ W_2 \end{pmatrix} x \right) = 2U_2 \, \vec{\sigma}(W_2 x) = \mathcal{N}_p\big(x; (2U_2, W_2)\big) \in V_{\sigma,p}$$

Furthermore, the linear combination of Φ_1 and Φ_2 goes through Φ. In particular, it holds that

$$\frac{1}{2}\Phi_1 + \frac{1}{2}\Phi_2 = \Phi.$$

Indeed, if we look at the function when linearly interpolating between $\hat{\theta}_1$ and $\hat{\theta}_2$, by Property 2.2, we get

$$\begin{aligned} \mathcal{N}_{2p}(\cdot; (1-\lambda)\hat{\theta}_1 + \lambda\hat{\theta}_2) &= (1-\lambda)\mathcal{N}_{2p}(\cdot; \hat{\theta}_1) + \lambda\mathcal{N}_{2p}(\cdot; \hat{\theta}_2) \\ &= (1-\lambda)\Phi_1 + \lambda\Phi_2 \end{aligned}$$

Substituting $\lambda = 1/2$ yields us

$$\frac{1}{2}\Phi_1 + \frac{1}{2}\Phi_2 = \mathcal{N}_{2p}\left(\cdot; \begin{pmatrix} U_1 & W_1 \\ U_2 & W_2 \end{pmatrix}\right) = \Phi \notin V_{\sigma,2p-1}.$$

According to property 2.1 we have $V_{\sigma,p} \subseteq V_{\sigma,p+1} \subseteq \ldots \subseteq V_{\sigma,2p-1}$, yielding us $\Phi \notin V_{\sigma,p}$ and non-convexity of the realization set. □

With a slight generalization of the above proof, we can identify another useful property of the set.

Property 2.4. For $i_1, \ldots, i_k \in \mathbb{N}_+$ let $\Phi_1 \in V_{\sigma,i_1}, \Phi_2 \in V_{\sigma,i_2}, \ldots, \Phi_k \in V_{\sigma,i_k}$. Furthermore, $p \coloneqq \sum_j i_j$. Then $\forall \lambda_1, \ldots, \lambda_k \in \mathbb{R}$, it holds that

$$\sum_{j=1}^{k} \lambda_j \Phi_j \in V_{\sigma,p}.$$

Furthermore, for all $\lambda_1, \ldots, \lambda_k \in \mathbb{R} \setminus \{0\}$, $\Phi \in V_{\sigma,p}$, and $k \leq p$ there exists $\Phi_1 \in V_{\sigma,i_1}, \Phi_2 \in V_{\sigma,i_2}, \ldots, \Phi_k \in V_{\sigma,i_k}$, with $i_1, \ldots, i_k \in \mathbb{N}_+$ and $\sum_j i_j = p$ such that the following is satisfied.

$$\Phi = \sum_{j=1}^{k} \lambda_j \Phi_j$$

Proof. The proof can be found in Appendix A. □

The above property is useful for our analysis and gives us multiple valuable results.

Corollary 2.2. Let $\Phi_1, \Phi_2 \in V_{\sigma,p}$, then their linear combination $\lambda\Phi_1 + (1-\lambda)\Phi_2$ is in $V_{\sigma,2p}$ for all $\lambda \in \mathbb{R}$.

Proof. This is a direct application of Property 2.4 with $k = 2$ and $i_1, i_2 = p$. □

This result is particularly useful to us combined with Property 2.2.

Corollary 2.3. For all $\Phi \in V_{\sigma,p}$ there exists $\{\Phi_i \in V_{\sigma,1}\}_{i\in[p]}$ such that Φ is the linear combination of the Φ_i-s. Furthermore, a linear combination of all $S \subseteq V_{\sigma,1}$ with $|S| = p$ is in $V_{\sigma,p}$.

Proof. This is a direct application of Property 2.4 with $k = p$ and $\forall j : i_j = 1$. □

Definition 2.1. Let V be a vector space and $X \subseteq V$. The convex hull $\mathrm{conv}(X) \subseteq V$ of X are all convex combinations of finitely many points in X.

$$\mathrm{conv}(X) \coloneqq \left\{ \sum_{i=1}^{k} \lambda_i x_i \;\middle|\; \begin{array}{l} k \in \mathbb{N}_+,\ x_1, \ldots, x_k \in X, \text{ and} \\ \lambda_1, \ldots, \lambda_k \in \mathbb{R}_{\geq 0} \text{ with } \sum_{i=1}^{k} \lambda_i = 1 \end{array} \right\}.$$

Property 2.5. We have that $\mathrm{conv}(V_{\sigma,1}) = V_{\sigma,\infty}$.

Proof. We prove this by double inclusion. First, we show that $\mathrm{conv}(V_{\sigma,1}) \subseteq V_{\sigma,\infty}$.

Indeed, let $\Phi \in \mathrm{conv}(V_{\sigma,1})$, by definition of the convex hull, Φ is a convex combination of finitely many, p, points in $V_{\sigma,1}$. By Corollary 2.3 we have that $\Phi \in V_{\sigma,p} \subseteq V_{\sigma,\infty}$. Hence, we have proved the first part, namely that $\mathrm{conv}(V_{\sigma,1}) \subseteq V_{\sigma,\infty}$.

For the other direction, take a $\Phi \in V_{\sigma,\infty}$ and denote by p the first width for which $\Phi \in V_{\sigma,p}$. By the definition of $V_{\sigma,\infty}$, this p exists and is finite. By Property 2.4 we have for suitably chosen λ_i-s that Φ is a convex combination of p many elements of $V_{\sigma,1}$. Hence, $\Phi \in \mathrm{conv}(V_{\sigma,1})$. □

While our set is non-convex, it does satisfy certain properties often considered in the convex geometry literature. Namely, that it is star convex.

Definition 2.2. Let V be a vector space. A set $S \subseteq V$ is star convex if there exists an $x_0 \in S$ such that for all $y \in S$ we have that $\forall \lambda \in [0, 1] : (1-\lambda)y + \lambda x_0 \in S$. Furthermore, any such x_0 is called the center of S.

Property 2.6. $V_{\sigma,p}$ is star convex, with the center being the origin.

Proof. Indeed, consider an arbitrary $\Phi \in V_{\sigma,p}$ represented by a $\theta = (U, W) \in \Theta_p$, that is, $\mathcal{N}_p(\cdot, \theta) = \Phi$. We have

$$(1 - \lambda)\Phi(x) = (1 - \lambda)U\vec{\sigma}(Wx) = \mathcal{N}_p(x; ((1 - \lambda)U, W))$$

And $\mathcal{N}_p(\cdot; ((1 - \lambda)U, W)) \in V_{\sigma,p}$. □

Due to star convexity we get that our realization set is path-connected.

Property 2.7. For all $p \in \mathbb{N}_+$ we have that the realization set $V_{\sigma,p}$ is path-connected.

Proof. Indeed, as $V_{\sigma,p}$ is star convex with center 0, for all $\Phi_1, \Phi_2 \in V_{\sigma,p}$ the path defined as

$$\gamma \colon [0, 1] \to V_{\sigma,p}, \quad \gamma(t) \coloneqq \begin{cases} (1 - 2t)\,\Phi_1 & \text{if } t \in [0, \frac{1}{2}], \\ (2t - 1)\,\Phi_2 & \text{if } t \in [\frac{1}{2}, 1], \end{cases}$$

connects Φ_1 to Φ_2 inside $V_{\sigma,p}$.

- $\gamma(0) = \Phi_1$ and $\gamma(1) = \Phi_2$.
- Both pieces are continuous and agree at $t = \frac{1}{2}$ (where both equal the zero realization $0 \in V_{\sigma,p}$), so γ is continuous on $[0, 1]$ by the pasting lemma.
- For $t \in [0, \frac{1}{2}]$, $\gamma(t) = (1 - 2t)\Phi_1 + 2t \cdot 0 \in V_{\sigma,p}$ by star convexity applied to $y = \Phi_1$ and $\lambda = 2t \in [0, 1]$. Symmetrically, $\gamma(t) \in V_{\sigma,p}$ for $t \in [\frac{1}{2}, 1]$.

Hence $V_{\sigma,p}$ is path-connected. □

Chapter 3

Results on the Disappearance of Spurious Valleys

In this chapter, we go over our main results with respect to the disappearance of spurious valleys. First, we consider arbitrary convex functional losses. Here we show positive and negative results on the disappearance of spurious valleys. We first show that by doubling the network size, we can escape previously spurious valleys. Then we prove that, in general, this doubling is a sharp bound. Finally, we strengthen our assumptions about the loss and look at the case where the functional loss is defined by the L^2 norm.

3.1 Arbitrary Convex Loss in Function Space

Theorem 3.1. Let Assumptions A1, A2, A3, and A4 hold, $p \in \mathbb{N}_+$, and $\theta^* \in \arg\min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$. For all $\theta \in \Theta_p$, and all $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$ there exists a path $\gamma : [0,1] \to \Theta_{2p}$ that is non-increasing with respect to $\mathcal{L}_{2p}$, satisfying $\gamma(0) = \hat{\theta}$ and $\gamma(1) \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p)$.

Proof. We construct a path satisfying the conditions of the theorem. Let $\theta = (U, W) \in \Theta_p$ be arbitrary, let $\theta^* = (U^*, W^*)$ be the global minimizer from the theorem statement, and let $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$. Set

$$\Phi_\theta \coloneqq \mathcal{N}_p(\cdot; \theta), \qquad \Phi_{\theta^*} \coloneqq \mathcal{N}_p(\cdot; \theta^*).$$

First segment. Define

$$\theta_1 \coloneqq \begin{pmatrix} U & W \\ 0 & W^* \end{pmatrix} \in \Theta_{2p}.$$

We have $\theta_1 \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$. Indeed, for all $x \in [0,1]^n$,

$$\mathcal{N}_{2p}(x; \theta_1) = \begin{pmatrix} U & 0 \end{pmatrix} \vec{\sigma}\left(\begin{pmatrix} W \\ W^* \end{pmatrix} x \right) = U\vec{\sigma}(Wx) = \mathcal{N}_p(x; \theta).$$

Since $\hat{\theta}, \theta_1 \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$, Assumption A3 gives a path $\gamma_1 : [0,1] \to \mathcal{F}_{\mathcal{N}}(\theta, 2p)$ with $\gamma_1(0) = \hat{\theta}$ and $\gamma_1(1) = \theta_1$. Along γ_1 the realized function, and hence $\mathcal{L}_{2p}$, is constant.

Second segment. Define

$$\theta_2 \coloneqq \begin{pmatrix} 0 & W \\ U^* & W^* \end{pmatrix} \in \Theta_{2p},$$

and let $\gamma_2 : [0,1] \to \Theta_{2p}$ be the linear interpolation $\gamma_2(t) \coloneqq (1-t)\theta_1 + t\theta_2$. The same computation as above gives $\mathcal{N}_{2p}(\cdot;\theta_2) = \mathcal{N}_p(\cdot;\theta^*) = \Phi_{\theta^*}$, so $\theta_2 \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p)$. Since θ_1 and θ_2 share the same weights within the activation function, Property 2.2 yields

$$\mathcal{N}_{2p}(\cdot;\gamma_2(t)) = (1-t)\Phi_\theta + t\Phi_{\theta^*}.$$

Define $f : [0,1] \to \mathbb{R}$ by $f(t) \coloneqq \mathcal{L}_{2p}(\gamma_2(t)) = l((1-t)\Phi_\theta + t\Phi_{\theta^*})$. We show f is non-increasing.

- f is convex on $[0,1]$ as the composition of the convex functional l (Assumption A2) with an affine map.
- f attains its minimum at $t = 1$: by Assumption A4, Φ_{θ^*} is a global minimizer of l over $V_{\sigma,\infty}$. Since $(1-t)\Phi_\theta + t\Phi_{\theta^*} \in V_{\sigma,2p} \subseteq V_{\sigma,\infty}$ for all $t \in [0,1]$, we obtain $f(t) \geq l(\Phi_{\theta^*}) = f(1)$.

A convex function on $[0,1]$ whose minimum is attained at the right endpoint is non-increasing on $[0,1]$, so $\mathcal{L}_{2p} \circ \gamma_2$ is non-increasing.

Gluing. By the pasting lemma applied to γ_1 and γ_2 (which agree at the join, both equaling θ_1), there exists a continuous path $\gamma : [0,1] \to \Theta_{2p}$ with $\gamma(0) = \hat{\theta}$ and $\gamma(1) = \theta_2 \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p)$. Since $\mathcal{L}_{2p} \circ \gamma_1$ is constant, $\mathcal{L}_{2p} \circ \gamma_2$ is non-increasing, and the two segments agree at the join, $\mathcal{L}_{2p} \circ \gamma$ is non-increasing on $[0,1]$. □

Corollary 3.1. Let the assumptions of Theorem 3.1 hold, and let $\mathcal{V} \subseteq \mathcal{P}(\Theta_{2p})$ denote the set of spurious valleys of $\mathcal{L}_{2p}$. Then for every $\theta \in \Theta_p$ and every $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$ we have that $\hat{\theta}$ is not in a spurious valley. That is, $\hat{\theta} \notin V$ for all $V \in \mathcal{V}$.

Proof. Let $\theta \in \Theta_p$ and $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$. By Theorem 3.1 there exists a path $\gamma : [0,1] \to \Theta_{2p}$, non-increasing with respect to $\mathcal{L}_{2p}$, with $\gamma(0) = \hat{\theta}$ and $\gamma(1) \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p)$. By Assumption A4, $\Phi_{\theta^*} = \mathcal{N}_p(\cdot;\theta^*)$ minimizes l over $V_{\sigma,\infty}$. Since $\min_{\vartheta \in \Theta_{2p}} \mathcal{L}_{2p}(\vartheta) = \min_{\Phi \in V_{\sigma,2p}} l(\Phi)$ and $\mathcal{N}_{2p}(\cdot;\gamma(1)) = \Phi_{\theta^*}$, the endpoint $\gamma(1)$ is a global minimizer of $\mathcal{L}_{2p}$. Therefore, $\hat{\theta}$ satisfies Property **P.1**, and Lemma 1.3 gives $\hat{\theta} \notin V$ for all $V \in \mathcal{V}$. □

With this we have shown that the affine variety we get via the extension of local minima into higher dimensions can be escaped if we double the number of neurons in our neural network. That is, we have given an answer to Question 1.1. Regardless of μ and in the convex functional loss l, any parameter $\theta \in \Theta_p$ is p-escapable. In Figure 3.1 we show a schematic of the theorem. In Figure 3.2 we show how walking around the functional equivalence class can decrease the barrier to a global minimizer.

The remainder of this section will focus on investigating whether doubling the network size is truly necessary for such a condition to occur. First, we show two examples where the linear combination can be done via fewer neurons and hence we can escape the functional equivalence class before we reach twice the number of neurons. Then we show that, in general, doubling the size is indeed necessary. Namely, we construct for all widths an adversarial loss, satisfying our current assumptions, and furthermore having the property that from $\mathcal{F}_{\mathcal{N}}(\theta, 2p-1)$ we cannot reach $\mathcal{F}_{\mathcal{N}}(\theta^*, 2p-1)$.

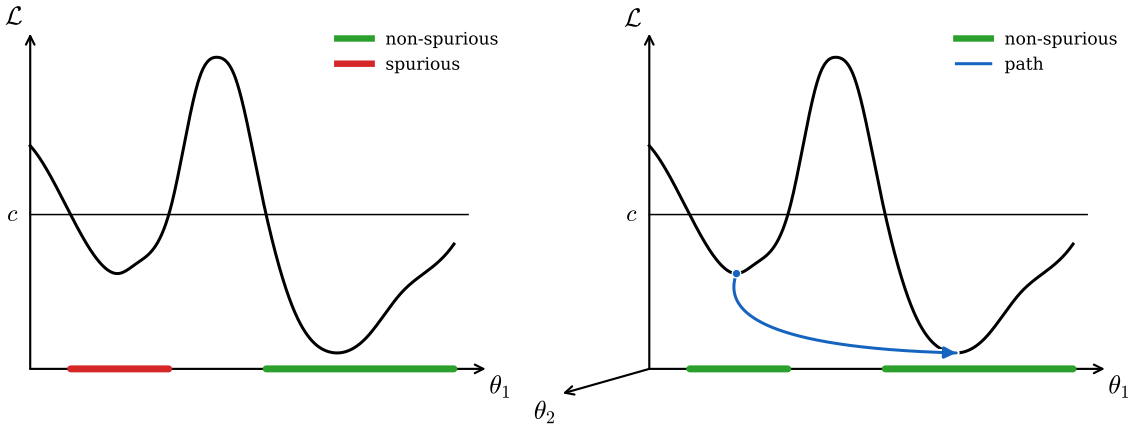


Figure 3.1: All spurious valleys become non-spurious when doubling the width of a neural network.

Examples. Two cases where fewer than $2p$ neurons suffice. In both, the construction mimics the proof of Theorem 3.1, so we omit details.

1. Let $\theta, \theta^* \in \Theta_p$ be such that $\theta \in \mathcal{F}_\mathcal{N}(\theta_{\min}, p)$ and $\theta^* \in \mathcal{F}_\mathcal{N}(\theta^*_{\min}, p)$ for minimal parameters $\theta_{\min} \in \Theta_{p_1}$ and $\theta^*_{\min} \in \Theta_{p_2}$, and set $q := p_1 + p_2$. Then for every $\hat{\theta} \in \mathcal{F}_\mathcal{N}(\theta, \max\{p, q\})$ there exists a non-increasing path $\gamma : [0,1] \to \Theta_{\max\{p,q\}}$ with respect to $\mathcal{L}_{\max\{p,q\}}$, satisfying $\gamma(0) = \hat{\theta}$ and $\gamma(1) \in \mathcal{F}_\mathcal{N}(\theta^*, \max\{p, q\})$.
2. Let $\theta = (U, W), \theta^* = (U^*, W^*) \in \Theta_p$ be such that k of the rows of W coincide with k rows of W^*. Then for every $\hat{\theta} \in \mathcal{F}_\mathcal{N}(\theta, 2p - k)$ there exists a non-increasing path $\gamma : [0,1] \to \Theta_{2p-k}$ with respect to $\mathcal{L}_{2p-k}$, satisfying $\gamma(0) = \hat{\theta}$ and $\gamma(1) \in \mathcal{F}_\mathcal{N}(\theta^*, 2p - k)$.

Next, we show our negative results. For all widths, p, we construct an adversarial loss such that with $p - 1$ neurons added, we cannot escape the functional equivalence class of two chosen parameters. This in particular implies that in the worst case loss we have $\varrho_p(p-1) = 0$ (Dirac deltas at the adversarial parameters), while still we have $\varrho_p(p) = 1$.

Theorem 3.2. Let Assumptions A1 and A5 hold and let $p \in \mathbb{N}_+$ and $M \in \mathbb{R}_+$. There exist $\theta, \theta^* \in \Theta_p$ and a convex functional loss $l : L^2([0,1]^n; \mathbb{R}^m) \to \mathbb{R}$ such that $\theta^* \in \arg\min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$ and for all $\hat{\theta} \in \mathcal{F}_\mathcal{N}(\theta, 2p-1)$ and all paths $\gamma : [0,1] \to \Theta_{2p-1}$ with $\gamma(0) = \hat{\theta}$ and $\gamma(1) \in \mathcal{F}_\mathcal{N}(\theta^*, 2p-1)$ there exists $t_0 \in [0,1]$ with $\mathcal{L}_{2p-1}(\gamma(t_0)) \geq M$.

Sketch of the proof. The proof relies on creating an adversarial loss. To this end, consider a function $\tilde{\Phi} \in V_{\sigma,2p}$ that for some $\epsilon > 0$ satisfies

$$\mathcal{B}(\tilde{\Phi}, \epsilon) \cap \overline{V_{\sigma,2p-1}} = \emptyset.$$

The above function $\tilde{\Phi}$ exists due to Assumption A5. Now, choose $\theta, \theta^* \in \Theta_p$ satisfying

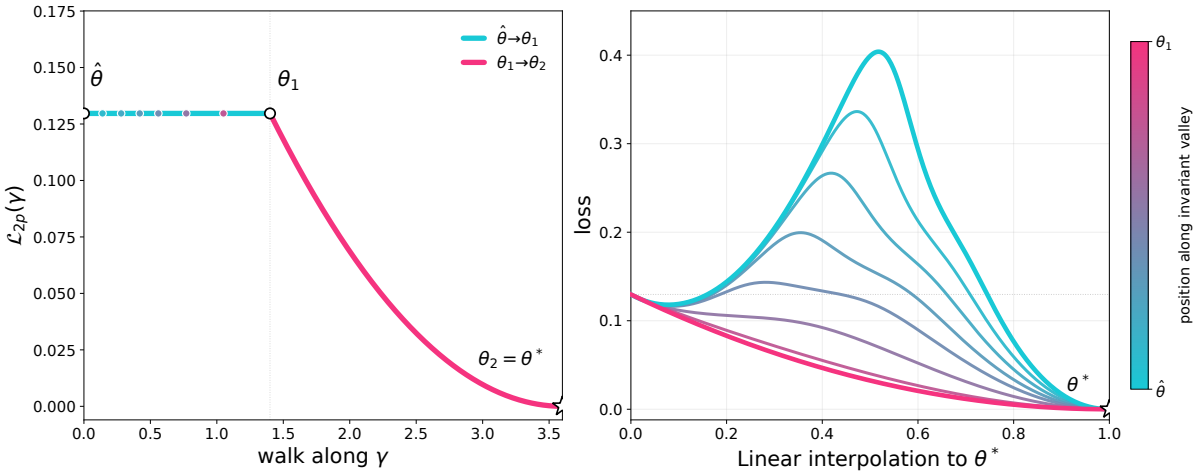


Figure 3.2: **Left.** A walk through (a) an invariance valley of a doubly over parameterized network and (b) descend into a global minimizer. θ_1, θ_2 is defined as in the proof of Theorem 3.1. **Right.** We sample points along the walk through the invariance valley from $\hat{\theta}$ to θ_1. At each sampled point we linearly interpolate to θ_2. The barrier disappears along the walk, until we reach θ_1 where a guaranteed decrease path exists.

$$\frac{1}{2}\mathcal{N}_p(\cdot;\theta) + \frac{1}{2}\mathcal{N}_p(\cdot;\theta^*) = \tilde{\Phi}.$$

Due to Property 2.4 there exist these θ, θ^* satisfying the above condition. Denote by $\Phi_1 = \mathcal{N}_p(\cdot;\theta)$ and $\Phi_2 = \mathcal{N}_p(\cdot;\theta^*)$. Furthermore, we define the set

$$S \coloneqq \{(1-t)\Phi_1 + t\Phi_2 : t \in [0,1]\}$$

Finally, for $c, d, e \in \mathbb{R}$ construct our adversarial loss as

$$l(\Phi) = c \inf_{z \in S}\{\|z - \Phi\|\} + d\langle(\Phi_1 - \Phi_2), \Phi\rangle + e$$

We claim that for suitable choices of c, d, e our theorem is satisfied. To give some insight as to why this works, consider this in the finite dimensional case shown in Figure 3.3. Our loss is constructed to create a valley between the two parameters with an arbitrarily sharp slope when deviating from it. Since we have a ball we cannot pass through in the middle of the valley, any path in $V_{\sigma,2p-1}$ cannot enter the ball, by driving the sharpness of the valley up, we have satisfied our theorem for any M. The full proof can be found in Appendix A.

Hence, we have shown that by assuming solely that the functional loss is convex, we generally need to double the width of a neural network to cause spurious valleys to disappear. In Figure 3.4 we show a schematic of this theorem.

Our loss construction is adversarial and, in general, would not hold for widely used loss functions. Therefore, in the next section, we look at a more widely used loss function, the mean squared error.

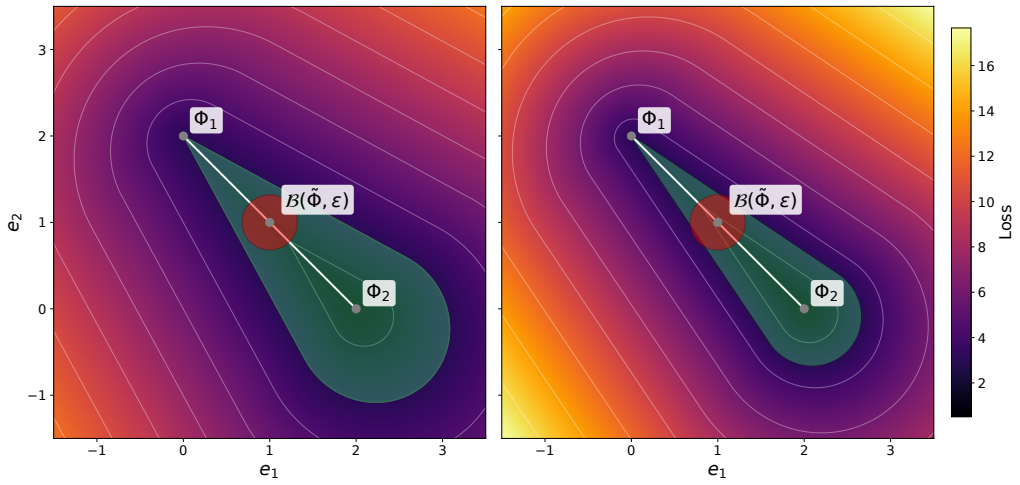


Figure 3.3: Figure of our adversarial loss function in two dimensions. The red ball denotes the functions that can only be realized via at least $2p$ neurons. The two functions Φ_1 and Φ_2 are shown as gray dots with the set S being the white line connecting the two. Green shows the loss values below the level of Φ_1. On the left we could theoretically have a path from Φ_1 to Φ_2, by increasing the sharpness of the loss any path taking on low functional loss must pass through the red ball.

3.2 L^2 Distance in Function Space

The following lemma holds for any convex functional loss, we will later use it with the squared L^2 loss.

Lemma 3.3. Let Assumptions A1, A2 and A3 hold, and let $\theta^* \in \Theta_p$ be such that

$$\mathcal{N}_p(\cdot; \theta^*) \in \arg\min_{\Phi \in V_{\sigma,p}} l(\Phi).$$

Then for all $\theta \in \mathcal{F}_{\mathcal{N}}((\mathbf{0}, \mathbf{0}), p)$ there exists a path $\gamma : [0,1] \to \Theta_p$ with $\gamma(0) = \theta$ and $\gamma(1) = \theta^*$ that is non-increasing with respect to $\mathcal{L}_p$.

Proof. Our proof revolves around constructing such a path.

First segment. Denote the weights of the global minimizer as $\theta^* = (U^*, W^*)$. Define the parameter

$$\theta_1 = (\mathbf{0}, W^*).$$

We have $\theta_1 \in \mathcal{F}_{\mathcal{N}}((\mathbf{0}, \mathbf{0}), p)$ as for all $x \in [0,1]^n$ we have $\mathbf{0}\vec{\sigma}(W^* x) = \mathbf{0}$. Since $\theta, \theta_1 \in \mathcal{F}_{\mathcal{N}}((\mathbf{0}, \mathbf{0}), p)$, Assumption A3 gives a path $\gamma_1 : [0,1] \to \mathcal{F}_{\mathcal{N}}((\mathbf{0}, \mathbf{0}), p)$ with $\gamma_1(0) = \theta$ and $\gamma_1(1) = \theta_1$. Along γ_1 the realized function is constant, hence $\mathcal{L}_p(\gamma_1(t))$ is constant for all $t \in [0,1]$.

Second segment. Now we consider the path $\gamma_2 : [0,1] \to \Theta_p$ defined by

$$\gamma_2(t) = (1-t)\theta_1 + t\theta^*.$$

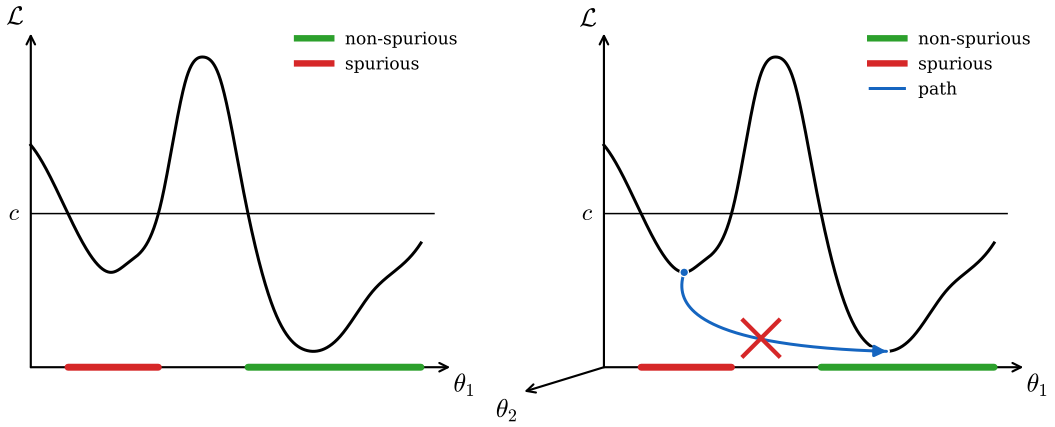


Figure 3.4: Unless we double the width of the neural network, arbitrarily bad spurious valleys may remain.

Since $\theta_1 = (\mathbf{0}, W^*)$ and $\theta^* = (U^*, W^*)$ share the same inner weight matrix W^*, Property 2.2 applies and gives

$$\mathcal{N}_p(\cdot\,; \gamma_2(t)) = (1-t)\cdot\mathbf{0} + t\,\mathcal{N}_p(\cdot\,; \theta^*).$$

Define $f(t) \coloneqq \mathcal{L}_p(\gamma_2(t)) = l(t\,\mathcal{N}_p(\cdot; \theta^*))$. This is convex on $[0,1]$ as the composition of the convex functional l with an affine map. Furthermore, $\mathcal{N}_p(\cdot; \theta^*)$ minimizes l over $V_{\sigma,p}$, so $f(t) \geq f(1)$ for all $t \in [0,1]$. A convex function on $[0,1]$ attaining its minimum at the right endpoint is non-increasing, and $\gamma_2(1) = \theta^*$.

Gluing. Apply the pasting lemma to γ_1 and γ_2, giving a path $\gamma : [0,1] \to \Theta_p$ with $\gamma(0) = \theta$ and $\gamma(1) = \theta^*$. Since $\mathcal{L}_p \circ \gamma_1$ is constant and $\mathcal{L}_p \circ \gamma_2$ is non-increasing, and the two segments agree at the join, $\mathcal{L}_p \circ \gamma$ is non-increasing on $[0,1]$. □

In the rest of this section, we examine the case where our functional loss is defined in terms of the L^2 norm with respect to some $f^* \in L^2([0,1]^n; \mathbb{R}^m)$. In particular,

$$l(\Phi) \coloneqq \|\Phi - f^*\|_{L^2}^2.$$

First, observe that the loss is convex in function space. The norm $\|\cdot\|_{L^2}$ is convex, and squaring a non-negative convex function yields a convex function, so l is convex. Therefore, Theorem 3.1 applies in this case as well.

Theorem 3.4. Let Assumption A1 hold, and let $l(\Phi) = \|\Phi - f^*\|_{L^2}^2$ for some $f^* \in L^2([0,1]^n; \mathbb{R}^m)$, and let $\theta^* = (U^*, W^*) \in \arg\min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$. Let μ be a probability measure on $(\Theta_p, B(\Theta_p))$ under which $U \overset{d}{=} -U$ and $U \perp\!\!\!\perp W$, where $\theta = (U, W) \sim \mu$. Then

$$\varrho_p(0) \geq \frac{1}{2}.$$

More precisely, every $\theta \in \Theta_p$ satisfying $\langle \mathcal{N}_p(\cdot; \theta), f^* \rangle_{L^2} \leq 0$ is 0-escapable via a non-increasing path consisting of three explicit linear segments:

$$(U, W) \;\to\; (0, W) \;\to\; (0, W^*) \;\to\; (U^*, W^*).$$

Sketch of the proof. Throughout this sketch, we adopt the standard shorthand for the measure:

$$\mu(f(\theta) \geq c) \coloneqq \mu(\{\theta \in \Theta_p : f(\theta) \geq c\}).$$

Consider the following walk in parameter space, consisting of three explicit linear segments. First, we linearly interpolate from $\theta = (U, W)$ to $\theta_1 = (0, W)$. Then, we linearly interpolate from $\theta_1 = (0, W)$ to $\theta_2 = (0, W^*)$. Finally, we linearly interpolate from $\theta_2 = (0, W^*)$ to $\theta^* = (U^*, W^*)$.

The middle segment $\theta_1 \to \theta_2$ realizes the function $\mathbf{0}$ throughout, since $\mathbf{0}\,\vec{\sigma}(((1-t)W + tW^*)x) = \mathbf{0}$ for all $t \in [0, 1]$, so the loss is constant along it. The final segment $\theta_2 \to \theta^*$ is non-increasing by the second-segment argument of Lemma 3.3 (both θ_2 and θ^* share the inner weight matrix W^*, and $\mathcal{N}_p(\cdot; \theta^*)$ minimizes l over $V_{\sigma,p}$). All that is left to show is that the first segment from θ to θ_1 is non-increasing with probability at least $1/2$.

We claim that the first segment is non-increasing whenever $\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0$, and we defer the proof of this to Appendix A. It therefore suffices to show

$$\mu\big(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0\big) \geq \frac{1}{2}.$$

We first condition on W. Since $U \perp\!\!\!\perp W$ and $U \overset{d}{=} -U$, the conditional law of U given W is symmetric, hence

$$\begin{aligned}
\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} &= -\langle U\vec{\sigma}(W\cdot), f^* \rangle_{L^2} \\
&\overset{d}{=} -\langle -U\vec{\sigma}(W\cdot), f^* \rangle_{L^2} \\
&= -\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2}.
\end{aligned}$$

We therefore have

$$\begin{aligned}
\mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0) &= \mu(-\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0) \\
&= \mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \leq 0) \\
&\geq \mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} < 0).
\end{aligned}$$

Since $\mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0)$ and $\mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} < 0)$ sum to one, the inequality above forces $\mu(\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0) \geq \frac{1}{2}$. In Appendix A we go into more detail on why the non-increasing condition holds when $\langle -\mathcal{N}_p(\cdot, \theta), f^* \rangle_{L^2} \geq 0$.

In Figure 3.5 we show a sketch of $L^2(\mathcal{U}; \mathbb{R}^m)$ and the necessary steps for the proof. In Figure 3.6 we show the path in parameter space and the loss along the path.

Remark 3.1. This probability can be greater than one half for two reasons. Firstly, it can be larger if we initialize W in the subspace where $\mathcal{N}_p(\cdot; \theta^*)$ can be found, since we do not need to walk to zero and can walk directly to the optimum. Secondly, we can continuously rotate W as well, which might yield a non-increasing path for some initializations.

Theorem 3.4 gives us $\varrho_p(0) \geq \frac{1}{2}$, and since ϱ_p is a probability it is trivially bounded above by 1, so $\varrho_p(0) \in [\frac{1}{2}, 1]$.

Property 3.1. ϱ_p is non-decreasing.

Indeed, for any parameter for which we can walk via k extra neurons, we can do the same walk using $k + 1$ extra neurons while keeping the last extra neuron all zeros.

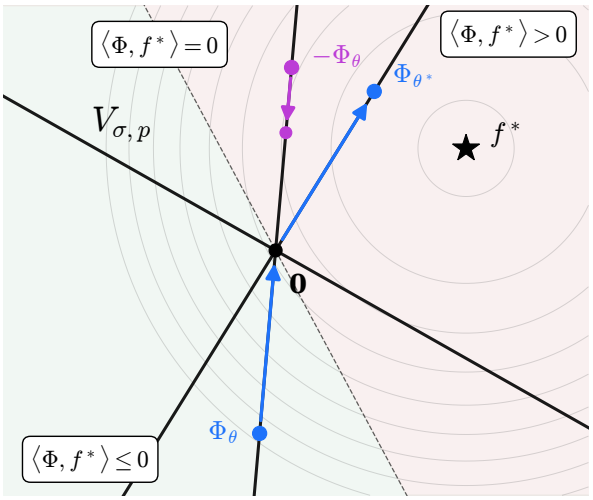


Figure 3.5: Schematic of the L^2 function space. Fixing W yields different linear subspaces in L^2, the union of which is $V_{\sigma,p}$. The space is split into two subsets, one with positive inner product with f^* and one with negative. We can walk to the origin and to the global minimizer in one case and cannot in the other.

Rotation of W. Our results on spurious valleys revolve around the exchange and introduction of new functions $\sigma(\langle w, \cdot \rangle)$ by moving the parameters U to zero and then extending them again. Another way of escaping to global minima might revolve around the continuous rotation of W. Indeed, for σ continuous, the map $\mathbb{R}^n \to L^2([0,1]^n), w \mapsto \sigma(\langle w, \cdot \rangle)$ is continuous. Hence, by moving w we rotate the function $\sigma(\langle w, \cdot \rangle)$ continuously in $L^2([0,1]^n)$. This can allow the existence of paths in the parameter-space that exploit this movement to reach a subspace, $\text{span}(\{\sigma(\langle w_i, \cdot \rangle)\}_{i \in [p]})$, containing the global minimizer. We demonstrate the following with an example.

Example. Consider $\theta^* = (2,3) \in \arg\min_{\vartheta \in \Theta_1} \mathcal{L}_1(\vartheta)$ and the starting parameter $\theta = (1,2) \in \Theta_1$ and let σ be the sigmoid activation function.
Consider the following paths:

- γ_1 moves first to $(0,2)$ then to $(0,3)$ and finally to θ^*
- γ_2 moves directly to θ^* by rotating w.

Figure 3.7 shows both paths. It can be seen that θ is not an element of a spurious valley as γ_2 moves directly to the global minimizer and is non-increasing with respect to the loss. Our current method, γ_1, does not take that path into account and there is a large increase in the loss value by only manipulating w when the u is zeroed out.

Restrictions on the Local Minimum. Another interesting finding is that strict local minima can only occur at projections in function space. Indeed, a local minimum implies that no parameter move strictly decreases the loss. If we fix some W, then as

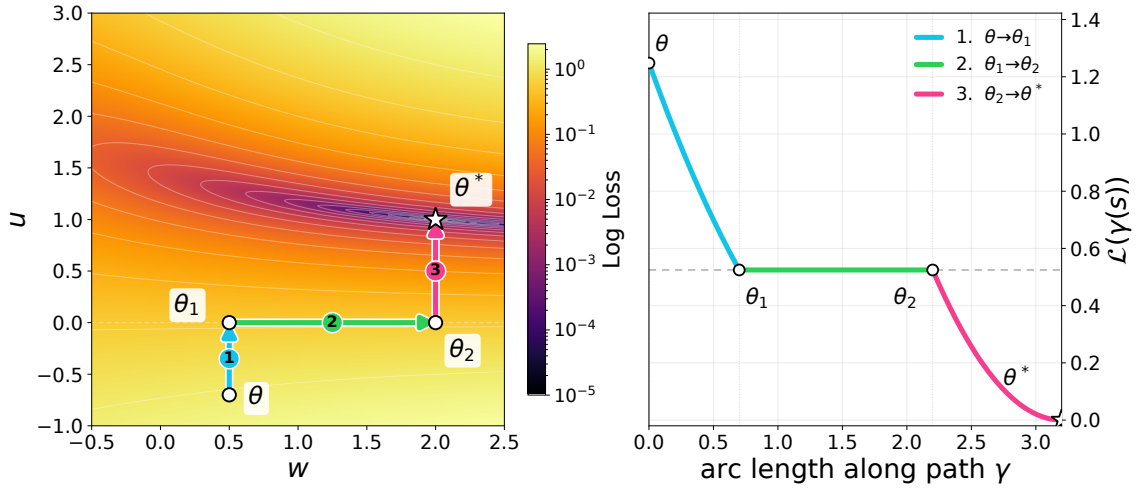


Figure 3.6: The three linear segments in parameter space, yielding the descending path to the global minimizer.

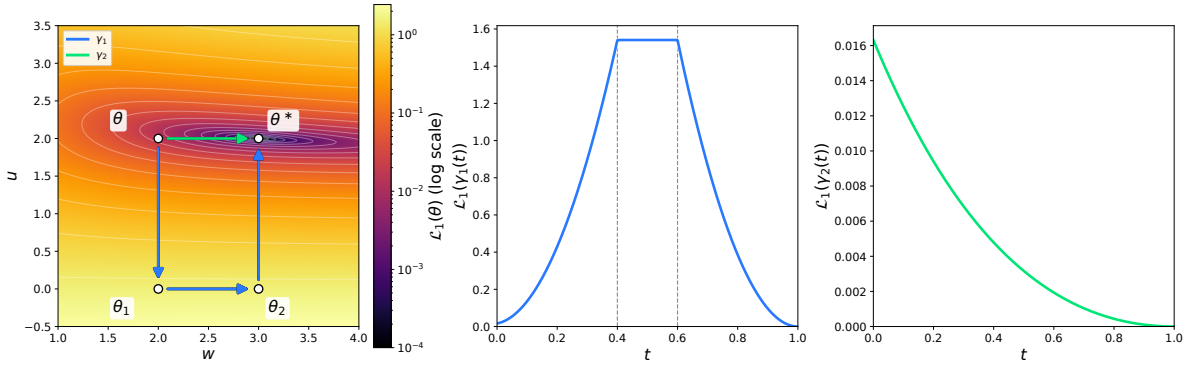


Figure 3.7: As seen in the figure, θ is not an element of a spurious valley. Yet along the path γ_1 the loss increases substantially. Conversely, if we consider the path γ_2 and move along w, i.e., rotate our function $\sigma(\langle w, \cdot \rangle)$ we can reach the global minimizer whilst non-increasing with respect to $\mathcal{L}_1$.

U ranges over $\mathbb{R}^{m \times p}$ the realized function $\mathcal{N}_p(\cdot; (U, W)) = \sum_{i=1}^{p} U_{:,i}\, \sigma(\langle w_i, \cdot \rangle)$ sweeps out the subspace

$$A_W := \left\{ \sum_{i=1}^{p} c_i\, \sigma(\langle w_i, \cdot \rangle) : c_1, \ldots, c_p \in \mathbb{R}^m \right\} \subseteq L^2(\mathcal{U}; \mathbb{R}^m),$$

where $w_i \in \mathbb{R}^n$ denotes the i-th row of W and $U_{:,i} \in \mathbb{R}^m$ the i-th column of U. Being finite dimensional, A_W is a closed subspace, so the orthogonal projection onto it is well defined. Consider the squared L^2 norm as before. For a given $\Phi \in A_W$ we define the error $e = \Phi - f^*$, and we denote by $\Pi_W(f^*)$ the orthogonal projection of f^* onto A_W. Setting $e_\parallel := \Phi - \Pi_W(f^*) \in A_W$ and $e_\perp := \Pi_W(f^*) - f^*$, the latter is orthogonal to A_W, so $e = e_\parallel + e_\perp$ is the decomposition of e into components parallel and perpendicular

to A_W. We can thus decompose the loss as

$$\begin{aligned} l(\Phi) &= \|\Phi - f^*\|_{L^2}^2 \\ &= \|\Phi - \Pi_W(f^*) + \Pi_W(f^*) - f^*\|_{L^2}^2 \\ &= \|e_\parallel\|_{L^2}^2 + 2\langle e_\parallel, e_\perp\rangle_{L^2} + \|e_\perp\|_{L^2}^2 \\ &= \|e_\parallel\|_{L^2}^2 + \|e_\perp\|_{L^2}^2, \end{aligned}$$

where the cross term vanishes since $e_\parallel \in A_W$ and $e_\perp \perp A_W$. As U varies with W fixed, only $e_\parallel = \Phi - \Pi_W(f^*)$ changes, while $e_\perp = \Pi_W(f^*) - f^*$ is determined by W alone. Hence if $\|e_\parallel\|_{L^2}^2 \neq 0$ we can move U so as to decrease the loss. Therefore, a local minimum in parameter space necessarily implements a function in $\{\Pi_W(f^*) : W \in \mathbb{R}^{p\times n}\}$, for otherwise the loss could be strictly decreased by moving U.

We end this chapter with an open question that we were unable to resolve.

Question 3.1. For L^2 norm squared loss is $\varrho_p(1) = 1$ for all μ probability measure on $(\Theta_p, B(\Theta_p))$?

By introducing one extra neuron, we gain flexibility to maneuver through the loss landscape, therefore we believe this case requires extra attention.

Chapter 4

Generalizations of Our Results

Our results can be generalized to some extent. In this chapter, we focus on detailing how these can be done. We go through possible generalizations paragraph by paragraph. Detailed proofs will be omitted, and focus will be placed on the intuition behind the ideas.

Assumption A3. Consider the initial question behind the problem. Our question was to see whether by extending the neural network with extra neurons we can escape from a parameter that is contained in a spurious valley. In order to study this, we introduced the notion of a functional equivalence class and looked at elements of it. Instead, by extending the neural network with all **0** neurons, we can skip the part of the first walk in most examples, where we walk along $\mathcal{F}_{\mathcal{N}}(\theta, k)$ for some k. Hence, our proofs simplify and many of our results hold even without Assumption A3 in a slightly different setting.

Assumption A4. This assumption was used for Theorem 3.1. Now, we show the technicalities needed to extend the results without the assumption holding.

Definition 4.1. Let $p \in \mathbb{N}_+$, $\epsilon \geq 0$ and $\mathcal{C}(S_c) \subseteq \mathcal{P}(\Theta_p)$ a set of path-connected components of a sub-level set $S_c \subseteq \Theta_p$ of the loss $\mathcal{L}_p$ with respect to $c \in \mathbb{R}$. The set of ϵ-spurious valleys, $\mathcal{V}_\epsilon \subseteq \mathcal{P}(\Theta_p)$ is defined as

$$\mathcal{V}_\epsilon := \{C \in \mathcal{C}(S_c) : C \cap \Theta_\epsilon^* = \emptyset\},$$

for $\Theta_\epsilon^* = \{\theta \in \Theta_p : \mathcal{L}_p(\theta) \leq \epsilon\} \subseteq \Theta_p$.

The useful property of convex functions is the following.

Proposition 4.1. Let X be a vector space, $f : X \to \mathbb{R}$ convex and continuous, and $x_1, x_2 \in X$ with $f(x_1) \geq f(x_2)$. Then there exists a $t_0 \in [0, 1]$ with

$$f((1 - t_0)x_1 + t_0 x_2) \leq f(x_2),$$

and for all $t \in [0, t_0]$ the map $t \mapsto f((1 - t)x_1 + tx_2)$ is non-increasing.

The proof is omitted for the sake of brevity. Now, a more general version of Theorem 3.1 can be stated informally as follows. At $p \in \mathbb{N}_+$ let $\epsilon = \min_{\vartheta \in \Theta_p} \mathcal{L}_p(\vartheta)$ and $\theta \in \Theta_p$ arbitrary. Then from any $\hat{\theta} \in \mathcal{F}_{\mathcal{N}}(\theta, 2p)$ there exists a non-increasing path to a $\theta^* \in \Theta_{2p}$ with $\mathcal{L}_{2p}(\theta^*) \leq \epsilon$. Hence, ϵ-spurious valleys are eliminated.

The proof essentially constructs the same path as in Theorem 3.1 then at the final line segment, the linear interpolation towards the global minimizer in Θ_p we invoke Proposition 4.1.

We remark that if a neural network can approximate sufficiently well (be that context dependent) for some width, then the generalized version of the theorem states that doubling the number of neurons connects parameters to sufficiently good solutions.

General Architectures. Our results generalize to different architectures. In particular, neural networks where the output map is linear.

Definition 4.2. A general neural network is a function

$$\mathcal{N} : [0,1]^n \times \mathbb{R}^{m \times p} \times \mathbb{R}^k \to \mathbb{R}^m$$

defined for $U \in \mathbb{R}^{m \times p}$, $w \in \mathbb{R}^k$ and a function $\varphi_p : \mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^p$ as

$$\mathcal{N}(x; (U, w)) = U\varphi_p(x; w).$$

Since most of our results used the output map to construct descent paths, many of them carry over directly to general neural networks.

1. **Theorem 3.1.** Assuming the architecture admits a block-additive composition $\varphi_{p+q}(x; (w_1, w_2)) = (\varphi_p(x; w_1); \varphi_q(x; w_2))$, the parameters $([U, 0], (w, w^*))$ and $([0, U^*], (w, w^*))$ realize $\mathcal{N}_p(\cdot; \theta)$ and $\mathcal{N}_p(\cdot; \theta^*)$ respectively, and the descent argument works here as well. Doubling the size of the inner block thus suffices to escape spurious valleys.

2. **Lemma 3.3.** The intermediate parameter $(0, w^*)$ realizes the zero function for any inner block φ, and the final segment keeps w^* fixed, so the linear interpolation in U remains linear in function space. The proof carries over without modification.

3. **Theorem 3.4 (L^2 probability bound).** The three-segment path uses only that $U = 0$ yields the zero realization for every w, the linearity in U, and the symmetry and independence of U under μ. None of these depend on the form of φ, so $\varrho_p(0) \geq \frac{1}{2}$ extends directly.

Chapter 5

Discussion

In this work, we have shown the following.

- Doubling the width of a two layered biasless neural network is sufficient to eliminate spurious valleys.
- Without further assumptions on the loss, this doubling is sharp.
- For the squared L^2 loss, under mild symmetry assumptions on the initialization that common schemes satisfy, at least half of all initializations lie outside any spurious valley, even without overparameterization.

Taken together, these results show that under sufficient overparameterization, spurious valleys no longer separate parameters from global minimizers and a non-increasing path between them always exists. We close by outlining several directions in which the analysis could be sharpened or extended.

Connection to optimization. Our results are purely geometric. We establish the existence of non-increasing paths to global minima, but say nothing about whether gradient-based methods can actually follow them. A natural next step is to investigate whether standard optimizers can leverage these paths, and, if not, whether new optimization methods can be designed to exploit the guarantees we provide.

Newly introduced spurious valleys. While overparameterization eliminates spurious valleys present in lower dimensional parameter spaces, the enlarged parameter space may also introduce new ones. Quantifying how many spurious valleys are created relative to the number eliminated would clarify the net effect of overparameterization on the loss landscape.

Tighter bounds via continuous movement of W. Our descent paths keep the inner weight matrix fixed on most segments and only manipulate U. As discussed in Chapter 3, continuously moving W rotates the feature functions $\sigma(\langle w, \cdot \rangle)$ in L^2, and exploiting this degree of freedom may yield tighter bounds, in particular reducing the amount of overparameterization required to escape spurious valleys.

Rate of disappearance of spurious valleys. Another question is how quickly $\varrho_p(k)$ approaches 1 for commonly used loss functions. If $\varrho_p(k) = 1$ is already attained at some k independent of p, one could develop tail bounds guaranteeing the absence of bad spurious valleys with high probability under sufficient overparameterization.

Bibliography

[1] K. Hornik, "Approximation capabilities of multilayer feedforward networks," *Neural networks*, vol. 4, no. 2, pp. 251–257, 1991.

[2] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Mathematics of control, signals and systems*, vol. 2, no. 4, pp. 303–314, 1989.

[3] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.

[4] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko *et al.*, "Highly accurate protein structure prediction with alphafold," *nature*, vol. 596, no. 7873, pp. 583–589, 2021.

[5] A. Novikov, N. Vũ, M. Eisenberger, E. Dupont, P.-S. Huang, A. Z. Wagner, S. Shirobokov, B. Kozlovskii, F. J. Ruiz, A. Mehrabian *et al.*, "Alphaevolve: A coding agent for scientific and algorithmic discovery," *arXiv preprint arXiv:2506.13131*, 2025.

[6] M. Belkin, D. Hsu, S. Ma, and S. Mandal, "Reconciling modern machine-learning practice and the classical bias–variance trade-off," *Proceedings of the National Academy of Sciences*, vol. 116, no. 32, pp. 15 849–15 854, 2019.

[7] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning requires rethinking generalization," *arXiv preprint arXiv:1611.03530*, 2016.

[8] V. N. Vapnik, *The Nature of Statistical Learning Theory*. New York: Springer, 1995.

[9] H. J. Sussmann, "Uniqueness of the weights for minimal feedforward nets with a given input-output map," *Neural networks*, vol. 5, no. 4, pp. 589–593, 1992.

[10] B. Simsek, F. Ged, A. Jacot, F. Spadaro, C. Hongler, W. Gerstner, and J. Brea, "Geometry of the loss landscape in overparameterized neural networks: Symmetries and invariances," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, M. Meila and T. Zhang, Eds., vol. 139. PMLR, 18–24 Jul 2021, pp. 9722–9732. [Online]. Available: https://proceedings.mlr.press/v139/simsek21a.html

[11] M. Farrugia-Roberts, "Functional equivalence and path connectivity of reducible hyperbolic tangent networks," in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. [Online]. Available: https://openreview.net/forum?id=ddKCg3OhGw

[12] V. Vlačić and H. Bölcskei, "Neural network identifiability for a family of sigmoidal nonlinearities," *Constructive Approximation*, vol. 55, no. 1, pp. 173–224, 2022.

[13] F. Albertini, E. Sontag, and V. Maillot, "Uniqueness of weights for neural networks," in *Artificial Neural Networks with Applications in Speech and Vision*, R. J. Mammone, Ed., 04 1993, pp. 113–125.

[14] L. Venturi, A. S. Bandeira, and J. Bruna, "Spurious valleys in one-hidden-layer neural network optimization landscapes," *Journal of Machine Learning Research*, vol. 20, no. 133, pp. 1–34, 2019. [Online]. Available: http://jmlr.org/papers/v20/18-674.html

[15] S. Fahlman and C. Lebiere, "The cascade-correlation learning architecture," *Advances in neural information processing systems*, vol. 2, 1989.

[16] M. Wynne-Jones, "Node splitting: A constructive algorithm for feed-forward neural networks," *Advances in neural information processing systems*, vol. 4, 1991.

[17] T. Chen, I. Goodfellow, and J. Shlens, "Net2net: Accelerating learning via knowledge transfer," *arXiv preprint arXiv:1511.05641*, 2015.

[18] L. Wu, D. Wang, and Q. Liu, "Splitting steepest descent for growing neural architectures," *Advances in neural information processing systems*, vol. 32, 2019.

[19] E. Kreyszig, *Introductory Functional Analysis with Applications*, ser. Wiley Classics Library. New York: John Wiley & Sons, 1989.

[20] M. Leshno, V. Y. Lin, A. Pinkus, and S. Schocken, "Multilayer feedforward networks with a nonpolynomial activation function can approximate any function," *Neural Networks*, vol. 6, no. 6, pp. 861–867, 1993. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608005801315

[21] C. Fefferman *et al.*, "Reconstructing a neural net from its output," *Revista Matemática Iberoamericana*, vol. 10, no. 3, pp. 507–556, 1994.

[22] M. Mondelli and A. Montanari, "On the connection between learning two-layer neural networks and tensor decomposition," in *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, K. Chaudhuri and M. Sugiyama, Eds., vol. 89. PMLR, 16–18 Apr 2019, pp. 1051–1060. [Online]. Available: https://proceedings.mlr.press/v89/mondelli19a.html

[23] J.-F. Le Gall, *Measure Theory, Probability, and Stochastic Processes*, ser. Graduate Texts in Mathematics. Springer Cham, 2022, vol. 295.

Appendix A

Proofs

Proof of Proposition 1.1

Proof. Fix $p \in \mathbb{N}_+$ and let $\Phi \in V_{\sigma,p}$ be arbitrary. By the definition of the realization set, there exists a parameter $\theta = (U, W) \in \Theta_p = \mathbb{R}^{m\times p} \times \mathbb{R}^{p\times n}$ such that

$$\Phi(x) = \mathcal{N}_p(x;\theta) = U\,\vec{\sigma}(Wx), \qquad \forall x \in [0,1]^n.$$

To show $\Phi \in L^2(\mathcal{U};\mathbb{R}^m)$, we verify that Φ is measurable and that $\int_{[0,1]^n} \|\Phi\|_2^2\, d\lambda < \infty$.

Measurability. Write $\Phi = B \circ \vec{\sigma} \circ A$ as the composition

$$[0,1]^n \xrightarrow{A} \mathbb{R}^p \xrightarrow{\vec{\sigma}} \mathbb{R}^p \xrightarrow{B} \mathbb{R}^m,$$

where $A(x) \coloneqq Wx$ and $B(y) \coloneqq Uy$. The maps A and B are linear maps between finite-dimensional Euclidean spaces and hence continuous. The map $\vec{\sigma} : \mathbb{R}^p \to \mathbb{R}^p$, $\vec{\sigma}(z) = (\sigma(z_1), \ldots, \sigma(z_p))^\top$, is continuous as well, as σ is continuous. A composition of continuous maps is continuous, hence Φ is continuous on $[0,1]^n$. By [23, Corollary 1.11] between two topological spaces equipped with their respective Borel σ-fields any continuous function is Borel measurable. Therefore, Φ is $B([0,1]^n)$-$B(\mathbb{R}^m)$-measurable.

Square integrability. Let $g \coloneqq \|\Phi(\cdot)\|_2 : [0,1]^n \to \mathbb{R}$. This is continuous as the composition of the continuous map Φ with the continuous norm $\|\cdot\|_2$. We know that g is bounded below by 0 as the norm is non negative. We proceed by showing that g is bounded above.

1. Local bounds from continuity.
   Fix $x \in [0,1]^n$. By continuity of g at x, applied with $\epsilon = 1$, there exists $\delta_x > 0$ such that the set
   $$U_x \coloneqq \{\, y \in [0,1]^n : \|y - x\|_2 < \delta_x \,\}$$
   satisfies $|g(y) - g(x)| < 1$, and hence $g(y) < g(x) + 1$, for all $y \in U_x$.

2. A finite subcover.
   Each U_x is relatively open in $[0,1]^n$ and contains x, so
   $$\bigcup_{x\in[0,1]^n} U_x$$

is an open cover of $[0,1]^n$. Since $[0,1]^n$ is compact, this cover admits a finite subcover. Hence, there exist finitely many points $x_1, \dots, x_k \in [0,1]^n$ with

$$[0,1]^n \subseteq \bigcup_{j=1}^{k} U_{x_j}.$$

3. A global bound.
   Let
   $$M := \max_{j\in[k]} \big(g(x_j) + 1\big) < \infty,$$
   be a maximum over finitely many real numbers. For any $x \in [0,1]^n$ there is an index j with $x \in U_{x_j}$, and by the construction of U_{x_j} we then have $g(x) < g(x_j) + 1 \le M$. Hence, together with non negativity, we have that $0 \le g(x) \le M$ for every $x \in [0,1]^n$.

Therefore we have

$$\int_{[0,1]^n} \|\Phi(x)\|_2^2 \, d\lambda(x) = \int_{[0,1]^n} g(x)^2 \, d\lambda(x) \le \int_{[0,1]^n} M^2 \, d\lambda(x) = M^2 \, \lambda([0,1]^n) = M^2 < \infty,$$

where we used $\lambda([0,1]^n) = 1$ and the bound on g we have developed.

Combining the two steps, Φ is measurable with $\int_{[0,1]^n} \|\Phi\|_2^2 \, d\lambda < \infty$, that is, $\Phi \in L^2(\mathcal{U}; \mathbb{R}^m)$. Since $\Phi \in V_{\sigma,p}$ was arbitrary, $V_{\sigma,p} \subseteq L^2(\mathcal{U}; \mathbb{R}^m)$. □

Proof of Lemma 1.3

Proof. Let $\theta \in \Theta_p$ satisfy Property **P.1**, and let $\gamma : [0,1] \to \Theta_p$ be a path as in **P.1**, namely $\gamma(0) = \theta$, $\gamma(1) \in \Theta^*$, and γ non-increasing with respect to $\mathcal{L}_p$.

Suppose, for the sake of contradiction, that $\theta \in V$ for some $V \in \mathcal{V}$. By the definition of $\mathcal{V}$, there exists $c \in \mathbb{R}$ such that $V \in \mathcal{C}(S_c)$ and $V \cap \Theta^* = \emptyset$. Since $V \subseteq S_c$, we have $\mathcal{L}_p(\theta) \le c$.

Because γ is non-increasing, for every $t \in [0,1]$ it holds that

$$\mathcal{L}_p(\gamma(t)) \le \mathcal{L}_p(\gamma(0)) = \mathcal{L}_p(\theta) \le c,$$

so $\gamma(t) \in S_c$. In particular, $\gamma([0,1]) \subseteq S_c$, and any two points $\gamma(s_1), \gamma(s_2) \in \gamma([0,1])$ are joined by the path $t \mapsto \gamma((1-t)s_1 + ts_2)$, whose image lies in $\gamma([0,1])$. Hence, any two points of $\gamma([0,1])$ are path connected within $\gamma([0,1])$.

We claim that any two points of $V \cup \gamma([0,1])$ are path connected within $V \cup \gamma([0,1])$. To this end, let $x, y \in V \cup \gamma([0,1])$. If $\{x, y\} \subseteq V$ they are path connected in V, and analogously if $\{x, y\} \subseteq \gamma([0,1])$. In the remaining case, without loss of generality, assume $x \in V$ and $y \in \gamma([0,1])$. Since $\theta \in V$ and V is path connected, there exists a path α from x to θ in V. Furthermore, as $\theta = \gamma(0) \in \gamma([0,1])$, there exists a path β from θ to y in $\gamma([0,1])$. The pasting lemma applied to α and β yields a path from x to y in $V \cup \gamma([0,1])$.

Therefore, $V \cup \gamma([0,1])$ is a subset of S_c in which any two points are connected by a path. Since V is maximal with this property and $V \subseteq V \cup \gamma([0,1])$, we conclude $V \cup \gamma([0,1]) = V$, that is, $\gamma([0,1]) \subseteq V$.

In particular $\gamma(1) \in V$. Combined with $\gamma(1) \in \Theta^*$, this gives $\gamma(1) \in V \cap \Theta^*$, contradicting $V \cap \Theta^* = \emptyset$. □

Proof of Property 2.4.

Proof. First we show that for $\Phi_1 \in V_{\sigma,i_1}, \Phi_2 \in V_{\sigma,i_2}, \ldots, \Phi_k \in V_{\sigma,i_k}$, we have that

$$\sum_{j=1}^{k} \lambda_j \Phi_j \in V_{\sigma,p}.$$

Since for $j \in [k]$ we have $\Phi_j \in V_{\sigma,i_j}$, denote by $\theta_j = (U_j, W_j) \in \Theta_{i_j}$ a parameter for which it holds that $\Phi_j = \mathcal{N}_{i_j}(\cdot, \theta_j)$. We now have that for arbitrary $\lambda_1, \ldots, \lambda_k$ and input x it holds that

$$\begin{aligned}\sum_{j=1}^{k} \lambda_j \Phi_j(x) &= \sum_{j=1}^{k} \lambda_j U_j \vec{\sigma}(W_j x) \\ &= \begin{pmatrix}\lambda_1 U_1 & \cdots & \lambda_k U_k\end{pmatrix} \vec{\sigma}\left(\begin{pmatrix} W_1 \\ \vdots \\ W_k \end{pmatrix} x\right) \in V_{\sigma,p}\end{aligned}$$

Conversely, consider a $\Phi \in V_{\sigma,p}$ and a partition of neurons $i_1, \ldots, i_k \in \mathbb{N}_+$ satisfying $\sum_j i_j = p$. Denote by $\theta = (\hat{U}, \hat{W}) \in \Theta_p$ a parameter for which $\Phi = \mathcal{N}_p(\cdot; \theta)$. The parameter θ can be partitioned according to

$$\theta = \begin{pmatrix} U_1 & W_1 \\ U_2 & W_2 \\ \vdots & \vdots \\ U_k & W_k \end{pmatrix},$$

for $U_j \in \mathbb{R}^{m \times i_j}$ and $W_j \in \mathbb{R}^{i_j \times n}$. Now consider the parameters defined by

$$\theta_1 = \begin{pmatrix} \frac{1}{\lambda_1} U_1 & W_1 \\ 0 & W_2 \\ \vdots & \vdots \\ 0 & W_k \end{pmatrix}, \quad \theta_2 = \begin{pmatrix} 0 & W_1 \\ \frac{1}{\lambda_2} U_2 & W_2 \\ \vdots & \vdots \\ 0 & W_k \end{pmatrix}, \ldots, \quad \theta_k = \begin{pmatrix} 0 & W_1 \\ 0 & W_2 \\ \vdots & \vdots \\ \frac{1}{\lambda_k} U_k & W_k \end{pmatrix}.$$

Similarly as in the proof of Property 2.3 for these parameters it holds that $\mathcal{N}_p(\cdot, \theta_j) \in V_{\sigma,i_j}$. Indeed, any of these functions can be just realized by the subset of neurons that are non-zero. By $\hat{U}_j$ denote parameters so that $\theta_j = (\hat{U}_j, \hat{W})$. Now, for their linear combination we have

$$\begin{aligned}\sum_{j=1}^{k} \lambda_j \Phi_j(x) &= \sum_{j=1}^{k} \lambda_j \mathcal{N}_p(x, \theta_j) \\ &= \sum_{j=1}^{k} \lambda_j \hat{U}_j \vec{\sigma}(\hat{W}x) \\ &= \left(\sum_{j=1}^{k} \lambda_j \hat{U}_j \right) \vec{\sigma}(\hat{W}x) \\ &= \hat{U}\vec{\sigma}(\hat{W}x) = \mathcal{N}_p(x, \theta) = \Phi(x)\end{aligned}$$

□

Proof of Theorem 3.2

Proof. Let $p \in \mathbb{N}_+$ and $M \in \mathbb{R}_+$. Throughout we abbreviate $\|\cdot\| = \|\cdot\|_{L^2}$ and $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{L^2}$.

Setting up the parameters. By Remark 2.1 applied at width $2p$, there exist $\tilde{\Phi} \in V_{\sigma,2p}$ and $\epsilon > 0$ such that

$$\mathcal{B}(\tilde{\Phi}, \epsilon) \cap \overline{V_{\sigma,2p-1}} = \emptyset. \tag{A.1}$$

Applying Property 2.4 to $\tilde{\Phi} \in V_{\sigma,2p}$ with $k = 2$, $i_1 = i_2 = p$, and $\lambda_1 = \lambda_2 = \frac{1}{2}$ yields $\Phi_1, \Phi_2 \in V_{\sigma,p}$ with

$$\tilde{\Phi} = \tfrac{1}{2}\Phi_1 + \tfrac{1}{2}\Phi_2.$$

Fix $\theta, \theta^* \in \Theta_p$ with $\mathcal{N}_p(\cdot; \theta) = \Phi_1$ and $\mathcal{N}_p(\cdot; \theta^*) = \Phi_2$. Note that $\Phi_1 \neq \Phi_2$, otherwise $\tilde{\Phi} = \Phi_1 \in V_{\sigma,p} \subseteq V_{\sigma,2p-1} \subseteq \overline{V_{\sigma,2p-1}}$, contradicting (A.1). Define

$$S \coloneqq \{(1-t)\Phi_1 + t\Phi_2 : t \in [0,1]\},$$

the segment from Φ_1 to Φ_2 in $L^2(\mathcal{U}; \mathbb{R}^m)$, of which $\tilde{\Phi}$ is the midpoint. Being the continuous image of $[0,1]$, the set S is compact, and being a line segment, it is convex.

Setting up the loss. As in the sketch, we work with the loss

$$l(\Phi) = c \inf_{z \in S} \|z - \Phi\| + d\langle \Phi_1 - \Phi_2, \Phi \rangle + e,$$

and determine $c, d, e \in \mathbb{R}$ such that:

(i) l is convex and continuous,

(ii) Φ_2 is the unique global minimizer of l, with $l(\Phi_2) = 0$,

(iii) $l(\Phi) > M$ whenever $\Phi - \tilde{\Phi} \perp \Phi_1 - \Phi_2$ and $\|\Phi - \tilde{\Phi}\| > \epsilon$.

The point of (iii) is that, as we will show, every path from θ to θ^* produces such a Φ on every admissible path.

Convexity and continuity. The distance $\Phi \mapsto \inf_{z\in S} \|z - \Phi\|$ to a convex set is convex and 1-Lipschitz, and $\Phi \mapsto d\langle \Phi_1 - \Phi_2, \Phi\rangle + e$ is affine. Hence l is convex and continuous regardless of c, d, e, giving (i).

Behavior on S. For any $z = (1-s)\Phi_1 + s\Phi_2 \in S$, the distance term is zero, and the loss can be expressed as

$$\begin{aligned} l(z) &= d\langle \Phi_1 - \Phi_2, (1-s)\Phi_1 + s\Phi_2\rangle + e \\ &= d(1-s)\langle \Phi_1 - \Phi_2, \Phi_1\rangle + ds\langle \Phi_1 - \Phi_2, \Phi_2\rangle + e \\ &= d(1-s)\langle \Phi_1 - \Phi_2, (\Phi_1 - \Phi_2) + \Phi_2\rangle + ds\langle \Phi_1 - \Phi_2, \Phi_2\rangle + e \\ &= d(1-s)\|\Phi_1 - \Phi_2\|^2 + d(1-s)\langle \Phi_1 - \Phi_2, \Phi_2\rangle + ds\langle \Phi_1 - \Phi_2, \Phi_2\rangle + e \\ &= d(1-s)\|\Phi_1 - \Phi_2\|^2 + d\langle \Phi_1 - \Phi_2, \Phi_2\rangle + e \\ &= d(1-s)\|\Phi_1 - \Phi_2\|^2 + \text{const}. \end{aligned}$$

Since $\|\Phi_1 - \Phi_2\|^2 > 0$, this function of s is minimized on $[0,1]$ at $s = 1$, i.e., at $z = \Phi_2$, precisely when $d > 0$. To enforce $l(\Phi_2) = 0$ we set

$$e = -d\langle \Phi_1 - \Phi_2, \Phi_2\rangle.$$

Under these two choices, $l(z) = d(1-s)\|\Phi_1 - \Phi_2\|^2 \geq 0$ for all $z \in S$, with equality iff $z = \Phi_2$.

Behavior off S. Since S is compact, for any $\Phi \in L^2(\mathcal{U}; \mathbb{R}^m)$ there exists $z^* \in S$ with $\|\Phi - z^*\| = \inf_{z\in S} \|z - \Phi\|$. Since $z^* \in S$, the distance term in $l(z^*)$ vanishes, so $l(z^*) = d\langle \Phi_1 - \Phi_2, z^*\rangle + e$. Using this together with $d > 0$ and the Cauchy-Schwarz inequality,

$$\begin{aligned} l(\Phi) &= c\|\Phi - z^*\| + d\langle \Phi_1 - \Phi_2, \Phi\rangle + e \\ &= c\|\Phi - z^*\| + d\langle \Phi_1 - \Phi_2, (\Phi - z^*) + z^*\rangle + e \\ &= c\|\Phi - z^*\| + d\langle \Phi_1 - \Phi_2, \Phi - z^*\rangle + d\langle \Phi_1 - \Phi_2, z^*\rangle + e \\ &= c\|\Phi - z^*\| + d\langle \Phi_1 - \Phi_2, \Phi - z^*\rangle + l(z^*) \\ &\geq c\|\Phi - z^*\| - d\|\Phi_1 - \Phi_2\|\,\|\Phi - z^*\| + l(z^*) \\ &= (c - d\|\Phi_1 - \Phi_2\|)\,\|\Phi - z^*\| + l(z^*). \end{aligned}$$

Combined with $l(z^*) \geq 0$ from the previous paragraph, we obtain $l(\Phi) \geq 0$ for all Φ as soon as

$$c \geq d\|\Phi_1 - \Phi_2\|.$$

Moreover, if we make the inequality strict, we obtain uniqueness of the global minimizer. For $\Phi \notin S$ we have $\|\Phi - z^*\| > 0$, so $l(\Phi) > l(z^*) \geq 0$, and on S, $l(z) > 0$ for $z \neq \Phi_2$ by the previous paragraph. Hence, Φ_2 is the unique Φ with $l(\Phi) = 0$, giving (ii). Lifting to parameters, $\mathcal{L}_p(\vartheta) = l(\mathcal{N}_p(\cdot;\vartheta)) \geq 0 = \mathcal{L}_p(\theta^*)$ for all $\vartheta \in \Theta_p$, with equality iff $\vartheta \in \mathcal{F}_{\mathcal{N}}(\theta^*, p)$.

Sharpness on the hyperplane through $\tilde{\Phi}$. Suppose $\Phi - \tilde{\Phi} \perp \Phi_1 - \Phi_2$ and $\|\Phi - \tilde{\Phi}\| > \epsilon$.

For any $z = (1-s)\Phi_1 + s\Phi_2 \in S$, we have

$$\begin{aligned}\tilde{\Phi} - z &= \frac{1}{2}\Phi_1 + \frac{1}{2}\Phi_2 - (1-s)\Phi_1 - s\Phi_2 \\ &= \left(\frac{1}{2} - 1 + s\right)\Phi_1 + \left(\frac{1}{2} - s\right)\Phi_2 \\ &= \left(s - \frac{1}{2}\right)\Phi_1 - \left(s - \frac{1}{2}\right)\Phi_2 \\ &= \left(s - \frac{1}{2}\right)(\Phi_1 - \Phi_2).\end{aligned}$$

This, together with orthogonality, yields

$$\begin{aligned}\|\Phi - z\|^2 &= \|(\Phi - \tilde{\Phi}) + (\tilde{\Phi} - z)\|^2 \\ &= \|\Phi - \tilde{\Phi}\|^2 + 2\langle \Phi - \tilde{\Phi}, \tilde{\Phi} - z\rangle + \|\tilde{\Phi} - z\|^2 \\ &= \|\Phi - \tilde{\Phi}\|^2 + 2(s - \tfrac{1}{2})\langle \Phi - \tilde{\Phi}, \Phi_1 - \Phi_2\rangle + (s - \tfrac{1}{2})^2\|\Phi_1 - \Phi_2\|^2 \\ &= \|\Phi - \tilde{\Phi}\|^2 + (s - \tfrac{1}{2})^2\|\Phi_1 - \Phi_2\|^2.\end{aligned}$$

This is minimized over $s \in [0,1]$ at $s = \frac{1}{2}$, where $z = \tilde{\Phi} \in S$, giving

$$\inf_{z \in S} \|z - \Phi\| = \|\Phi - \tilde{\Phi}\| > \epsilon.$$

For the linear term, using $\tilde{\Phi} - \Phi_2 = \frac{1}{2}(\Phi_1 - \Phi_2)$ and again the orthogonality,

$$\begin{aligned}\langle \Phi_1 - \Phi_2, \Phi\rangle &= \langle \Phi_1 - \Phi_2, \tilde{\Phi} + (\Phi - \tilde{\Phi})\rangle \\ &= \langle \Phi_1 - \Phi_2, \tilde{\Phi}\rangle + \langle \Phi_1 - \Phi_2, \Phi - \tilde{\Phi}\rangle \\ &= \langle \Phi_1 - \Phi_2, \tilde{\Phi}\rangle \\ &= \langle \Phi_1 - \Phi_2, \Phi_2 + (\tilde{\Phi} - \Phi_2)\rangle \\ &= \langle \Phi_1 - \Phi_2, \Phi_2\rangle + \langle \Phi_1 - \Phi_2, \tilde{\Phi} - \Phi_2\rangle \\ &= \langle \Phi_1 - \Phi_2, \Phi_2\rangle + \langle \Phi_1 - \Phi_2, (1/2)\Phi_1 + (1/2)\Phi_2 - \Phi_2\rangle \\ &= \langle \Phi_1 - \Phi_2, \Phi_2\rangle + \frac{1}{2}\langle \Phi_1 - \Phi_2, \Phi_1 - \Phi_2\rangle \\ &= \langle \Phi_1 - \Phi_2, \Phi_2\rangle + \frac{1}{2}\|\Phi_1 - \Phi_2\|^2,\end{aligned}$$

Combined with $e = -d\langle \Phi_1 - \Phi_2, \Phi_2\rangle$, this gives

$$d\langle \Phi_1 - \Phi_2, \Phi\rangle + e = \tfrac{d}{2}\|\Phi_1 - \Phi_2\|^2.$$

Substituting both into the definition of l,

$$\begin{aligned}l(\Phi) &= c\|\Phi - \tilde{\Phi}\| + \tfrac{d}{2}\|\Phi_1 - \Phi_2\|^2 \\ &> c\epsilon + \tfrac{d}{2}\|\Phi_1 - \Phi_2\|^2.\end{aligned}$$

Hence (iii) holds as soon as

$$c\epsilon + \tfrac{d}{2}\|\Phi_1 - \Phi_2\|^2 \geq M.$$

Choice of constants. The three conditions

$$d > 0, \qquad c > d\|\Phi_1 - \Phi_2\|, \qquad c\epsilon + \tfrac{d}{2}\|\Phi_1 - \Phi_2\|^2 \geq M$$

are jointly satisfied by, for example:

$$d = 1, \qquad c = \|\Phi_1 - \Phi_2\| + \frac{M}{\epsilon}, \qquad e = -\langle \Phi_1 - \Phi_2, \Phi_2 \rangle.$$

Indeed, $d = 1 > 0$; $c = \|\Phi_1 - \Phi_2\| + M/\epsilon > \|\Phi_1 - \Phi_2\| = d\|\Phi_1 - \Phi_2\|$; and

$$c\epsilon + \tfrac{1}{2}\|\Phi_1 - \Phi_2\|^2 = \epsilon\|\Phi_1 - \Phi_2\| + M + \tfrac{1}{2}\|\Phi_1 - \Phi_2\|^2 \geq M.$$

Hence (i), (ii), and (iii) all hold for this choice.

Continuity of the realization map. Let $\Psi : \Theta_{2p-1} \to L^2(\mathcal{U}; \mathbb{R}^m)$, $\Psi(\vartheta) \coloneqq \mathcal{N}_{2p-1}(\cdot; \vartheta)$. For σ continuous, the map $\mathbb{R}^n \to L^2(\mathcal{U})$, $w \mapsto \sigma(\langle w, \cdot \rangle)$, is continuous. Writing $\vartheta = (U, W)$ with columns $U_{:,j} \in \mathbb{R}^m$ and rows $W_{j,:} \in \mathbb{R}^n$,

$$\Psi(\vartheta) = \sum_{j=1}^{2p-1} U_{:,j}\, \sigma(\langle W_{j,:}, \cdot \rangle).$$

The coordinate maps $\vartheta \mapsto U_{:,j}$ and $\vartheta \mapsto W_{j,:}$ are continuous, and scalar multiplication and addition on $L^2(\mathcal{U}; \mathbb{R}^m)$ are continuous, so Ψ is continuous as a finite sum of products of continuous maps.

Every admissible path attains loss at least M. Let $\gamma : [0, 1] \to \Theta_{2p-1}$ be any path with $\gamma(0) \in \mathcal{F}_{\mathcal{N}}(\theta, 2p-1)$ and $\gamma(1) \in \mathcal{F}_{\mathcal{N}}(\theta^*, 2p-1)$, and define

$$\Gamma : [0, 1] \to L^2(\mathcal{U}; \mathbb{R}^m), \qquad \Gamma(t) \coloneqq \mathcal{N}_{2p-1}(\cdot; \gamma(t)) = \Psi(\gamma(t)).$$

Since Ψ is continuous and γ is a path, $\Gamma = \Psi \circ \gamma$ is a continuous path in $V_{\sigma,2p-1}$ with $\Gamma(0) = \Phi_1$ and $\Gamma(1) = \Phi_2$.

Consider the continuous function $h : [0, 1] \to \mathbb{R}$ defined by

$$h(t) \coloneqq \langle \Gamma(t) - \tilde{\Phi}, \Phi_1 - \Phi_2 \rangle.$$

At the endpoints, using $\Phi_1 - \tilde{\Phi} = \frac{1}{2}(\Phi_1 - \Phi_2)$ and $\Phi_2 - \tilde{\Phi} = -\frac{1}{2}(\Phi_1 - \Phi_2)$,

$$h(0) = \langle \Phi_1 - \tilde{\Phi}, \Phi_1 - \Phi_2 \rangle = \tfrac{1}{2}\|\Phi_1 - \Phi_2\|^2 > 0,$$
$$h(1) = \langle \Phi_2 - \tilde{\Phi}, \Phi_1 - \Phi_2 \rangle = -\tfrac{1}{2}\|\Phi_1 - \Phi_2\|^2 < 0.$$

By the intermediate value theorem, there exists $t_0 \in [0, 1]$ with $h(t_0) = 0$, i.e., $\Gamma(t_0) - \tilde{\Phi} \perp \Phi_1 - \Phi_2$.

Since $\Gamma(t_0) \in V_{\sigma,2p-1} \subseteq \overline{V_{\sigma,2p-1}}$, (A.1) forces $\Gamma(t_0) \notin \mathcal{B}(\tilde{\Phi}, \epsilon)$, so $\|\Gamma(t_0) - \tilde{\Phi}\| > \epsilon$. Applying condition (iii) on l,

$$\mathcal{L}_{2p-1}(\gamma(t_0)) = l(\Gamma(t_0)) > M.$$

Since γ was arbitrary, the claim follows. □

Proof of Theorem 3.4

Proof. We prove the claim deferred from the main text. The first segment of the path, the linear interpolation from $\theta = (U, W)$ to $\theta_1 = (0, W)$, is non-increasing with respect to $\mathcal{L}_p$ whenever $\langle -\mathcal{N}_p(\cdot; \theta), f^* \rangle_{L^2} \geq 0$.

Write $\Phi \coloneqq \mathcal{N}_p(\cdot; \theta)$. Along the first segment, at parameter $s \in [0, 1]$ the parameter is $((1 - s)U, W)$, which realizes

$$\mathcal{N}_p(\cdot; ((1 - s)U, W)) = (1 - s)U\, \vec{\sigma}(W\cdot) = (1 - s)\Phi.$$

Hence the loss along the segment is

$$g(s) \coloneqq \mathcal{L}_p(((1-s)U, W)) = \|(1-s)\Phi - f^*\|_{L^2}^2 = (1-s)^2\|\Phi\|_{L^2}^2 - 2(1-s)\langle \Phi, f^* \rangle_{L^2} + \|f^*\|_{L^2}^2,$$

which is differentiable in s with

$$g'(s) = -2(1 - s)\|\Phi\|_{L^2}^2 + 2\langle \Phi, f^* \rangle_{L^2}.$$

For every $s \in [0, 1]$ we have $1 - s \geq 0$ and $\|\Phi\|_{L^2}^2 \geq 0$, so the first term of $g'(s)$ is nonpositive. The second term is nonpositive by assumption. Thus $g'(s) \leq 0$ on $[0, 1]$, so g is non-increasing and the first segment is non-increasing with respect to $\mathcal{L}_p$.

This is the claim deferred from the main text. Together with the middle segment $(0, W) \to (0, W^*)$, which realizes the constant function $\mathbf{0}$ and hence has constant loss, and the final segment $(0, W^*) \to (U^*, W^*)$, which is non-increasing by the second-segment argument of Lemma 3.3, the pasting lemma yields a non-increasing path from θ to θ^*. □

Appendix B

Mathematical Preliminaries

The following introduction to the mathematical background draws on the contents of two ETH Zurich courses: Mathematics of Information, offered by the Chair for Mathematical Information Science, and Probability Theory, offered by the Probability Theory Group. In addition, it draws on [19, 23].

Definition B.1. Let X be a set. A metric on X is a function $d : X \times X \to \mathbb{R}$ that satisfies the following properties for all $x, y, z \in X$.

- Positivity: $d(x, y) \geq 0$ and $d(x, y) = 0 \Leftrightarrow x = y$.
- Symmetry: $d(x, y) = d(y, x)$.
- Triangle Inequality: $d(x, z) \leq d(x, y) + d(y, z)$.

Definition B.2. Let X be a set and d a metric on X. The pair (X, d) is called a metric space.

Definition B.3. Let (X, d) be a metric space. A set $S \subseteq X$ is called an open set if for all $x_0 \in S$ there exists an $\epsilon > 0$ such that for all $x \in X$ with $d(x, x_0) < \epsilon$ we have $x \in S$.

Definition B.4. Let (X, d) be a metric space. A sequence $(x_i)_{i \in \mathbb{N}}$ of elements of X converges to x if for all $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that for all $i \geq N$ it holds that

$$d(x, x_i) \leq \epsilon$$

Definition B.5. Let (X, d) be a metric space. A sequence $(x_i)_{i \in \mathbb{N}}$ of elements of X is called a Cauchy sequence if for all $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that if $i, j \geq N$

$$d(x_i, x_j) \leq \epsilon$$

Definition B.6. A metric space (X, d) is complete if every Cauchy sequence in X converges to an element of X.

Definition B.7. Let (X, d) be a metric space. A subset $S \subseteq X$ is called a closed set if every sequence $(x_i)_{i \in \mathbb{N}}$ of elements of S that converges to an element of X converges to an element of S. We write $\overline{S}$ for the smallest closed subset of X containing S.

Definition B.8. Let $A \subseteq B$ be subsets of a metric space (X, d). We say that A is dense in B if $\overline{A} = B$.

Examples.

1. $(0,1) \coloneqq \{x \in \mathbb{R} : 0 < x < 1\} \subseteq \mathbb{R}$ is an open set.
2. $[0,1] \coloneqq \{x \in \mathbb{R} : 0 \leq x \leq 1\} \subseteq \mathbb{R}$ is a closed set.
3. $\mathbb{R}$ and $\emptyset$ are both closed and open subsets of $\mathbb{R}$.
4. $[0,1) \coloneqq \{x \in \mathbb{R} : 0 \leq x < 1\} \subseteq \mathbb{R}$ is neither closed nor open.

Definition B.9. Let (X, d_X) and (Y, d_Y) be two metric spaces. We say that the function $f : X \to Y$ is continuous at a point $x_0 \in X$ if for all $\epsilon > 0$ there exists a $\delta > 0$ such that for all $x \in X$ with $d_X(x, x_0) < \delta$ we have

$$d_Y(f(x), f(x_0)) < \epsilon$$

Definition B.10. A set X with two operations $(+, \cdot)$ is a vector space over the field $\mathbb{R}$ if the following hold:

- The first operation, called vector addition, $+ : X \times X \to X$, satisfies
  - For all $x, y, z \in X$ we have $(x + y) + z = x + (y + z)$.
  - For all $x, y \in X$ it holds that $x + y = y + x$.
- There exists an element $0 \in X$, called the zero vector, that for all $x \in X$ satisfies $x + 0 = x$.
- For all $x \in X$ there exists an element $(-x) \in X$ that satisfies $x + (-x) = 0$.
- The second operation, called scalar multiplication, $\cdot : \mathbb{R} \times X \to X$ satisfies
  - For all $x \in X$ it holds that $1 \cdot x = x$.
  - For all $a, b \in \mathbb{R}$ and $x \in X$ it holds that $a \cdot (b \cdot x) = (ab) \cdot x$.
  - For all $a, b \in \mathbb{R}$ and $x \in X$ we have $(a + b) \cdot x = a \cdot x + b \cdot x$
  - For all $a \in \mathbb{R}$ and $x, y \in X$ it holds that $a \cdot (x + y) = a \cdot x + a \cdot y$.

Examples.

1. The set $\mathbb{R}^n \coloneqq \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\}$ with operators for $x, y \in \mathbb{R}^n$ and $c \in \mathbb{R}$ defined as

   $$x + y = (x_1 + y_1, \ldots, x_n + y_n) \qquad c \cdot x = (cx_1, \ldots, cx_n)$$

   forms a vector space.
2. The set of all functions, $\mathcal{F}(\mathbb{R}^n, \mathbb{R}^m)$, mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$ forms a vector space under pointwise addition and multiplication.

Definition B.11. Let X be a vector space over $\mathbb{R}$. A finite set of vectors $x_1, \ldots, x_k \in X$ are independent if for all $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$

$$\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_k x_k = 0$$

implies that $\lambda_1, \ldots, \lambda_k = 0$. An infinite set S of vectors in X is said to be independent if any finite subset of S is linearly independent. A set S of vectors in X is linearly dependent if it is not linearly independent.

Examples.

1. In $\mathbb{R}^2$ the vectors $\{(0,1),(1,0)\}$ are linearly independent.
2. Any set S of vectors in X, containing the zero vector, is linearly dependent.

Definition B.12. A vector space X is N-dimensional if there exists a set of vectors S in X with cardinality N that are linearly independent, but any set of vectors of cardinality $N+1$ is linearly dependent. If for some $N \in \mathbb{N}$ the vector space is N-dimensional, it is called a finite dimensional vector space. Otherwise, it is infinite dimensional.

Examples.

1. The vector space $\mathbb{R}^n$ is n-dimensional.
2. The set of all functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ is infinite dimensional.

Definition B.13. Let X be a vector space over $\mathbb{R}$. A norm $\|\cdot\|$ on X is a function that maps elements from X to $\mathbb{R}$ and satisfies the following properties.

- Positivity: For all $x \in X$ we have $\|x\| \geq 0$ and $\|x\| = 0$ implies $x = 0$.
- Positive homogeneity: For all $x \in X$ and $c \in \mathbb{R}$ we have $\|cx\| = |c|\|x\|$.
- Triangle inequality: For all $x, y \in X$ it holds that $\|x+y\| \leq \|x\| + \|y\|$

Examples.

1. The L^p norm $\|\cdot\|_p$ on $\mathbb{R}^n$ is defined as

$$\|x\|_p \coloneqq \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$$

2. The L^∞ norm $\|\cdot\|_\infty$ on $\mathbb{R}^n$ is defined as

$$\|x\|_\infty \coloneqq \max\{|x_1|, \ldots, |x_n|\}$$

Definition B.14. Let X be a vector space equipped with a norm $\|\cdot\|$. The pair $(X, \|\cdot\|)$ is called a normed vector space.

Remark B.1. Let $(X, \|\cdot\|)$ be a normed vector space. Then for $x, y \in X$ the function $d(x, y) = \|x - y\|$ is a metric. Hence, every normed space is a metric space.

Definition B.15. Let X be a vector space over $\mathbb{R}$. An inner product over $\mathbb{R}$ is a function $X \times X \to \mathbb{R}$, denoted by $\langle \cdot, \cdot \rangle$, such that for all $x, y, z \in X$ and $c \in \mathbb{R}$ the following are satisfied.

- Symmetry: $\langle x, y \rangle = \langle y, x \rangle$.
- Bilinearity: $\langle cx, y \rangle = c\langle x, y \rangle$
- Bilinearity: $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$
- Positivity: $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0$ implies $x = 0$.

Example.

1. The dot product $\langle \cdot, \cdot \rangle_{\mathbb{R}^m}$ on $\mathbb{R}^n$ defined for $x, y \in X$ by

$$\langle x, y \rangle_{\mathbb{R}^m} \coloneqq \sum_{i=1}^{n} x_i y_i$$

   is an inner product on X.

Definition B.16. Let X be a vector space equipped with an inner product $\langle \cdot, \cdot \rangle$. Then the pair $(X, \langle \cdot, \cdot \rangle)$ is called an inner product space.

Remark B.2. If $\langle \cdot, \cdot \rangle$ is an inner product on X, then $\|x\| \coloneqq \sqrt{\langle x, x \rangle}$ is a norm on X. Hence, any inner product space is a normed space.

Definition B.17. A Banach space is a complete normed space.

Definition B.18. A Hilbert space is a complete inner product space.

We now turn our attention to concepts of probability and measure theory underlying this work.

Definition B.19. Let Ω be a set and denote by $\mathcal{P}(\Omega)$ the power set of Ω, i.e. the set of every subset of Ω. The set $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is called a σ-field on Ω if the following are satisfied:

- $\Omega \in \mathcal{A}$
- $\mathcal{A}$ is closed under complement: $A \in \mathcal{A} \Rightarrow \Omega \setminus A \in \mathcal{A}$
- $\mathcal{A}$ is closed under countable unions: If $(A_i)_{i=1}^{\infty}$ is a sequence of elements in $\mathcal{A}$ then so is $\bigcup_{i \geq 1} A_i$

Examples.

1. Let Ω be a set. The trivial σ-field, defined as $\mathcal{A} = \{\emptyset, \Omega\}$, is the smallest σ-field on Ω.
2. $\mathcal{P}(\Omega)$, the set of all sets of Ω, is the largest σ-field on Ω.
3. Consider $\Omega = \{a, b, c\}$, the set $\mathcal{A} = \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\}$ is a σ-field on Ω.
4. For the set $\Omega = \{a, b, c\}$, the set $\mathcal{A} = \{\emptyset, \{a\}, \{b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$ is not a σ-field. It satisfies $\Omega \in \mathcal{A}$ and closure under complement, but is not closed under countable unions as $\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{A}$.

As illustrated by the last example, σ-fields are hard to construct. Given a set $M \subseteq \mathcal{P}(\Omega)$ we often want to consider a σ-field containing M, to be able to measure elements of M. To explicitly construct this set would require iteratively checking if the conditions of σ-fields are satisfied, and adding elements to the set M. Our next definition allows us to work with generated σ-fields.

Definition B.20. Let Ω be a set, and $M \subseteq \mathcal{P}(\Omega)$. The σ-field generated by M is defined as

$$\sigma(M) \coloneqq \bigcap_{\substack{\mathcal{A} \text{ } \sigma\text{-field on } \Omega \\ M \subseteq \mathcal{A}}} \mathcal{A}.$$

It is the smallest σ-field containing M.

Examples.

1. Let Ω be a set, $\mathcal{A}$ a σ-field on Ω. The σ-field generated by $\mathcal{A}$ is $\sigma(\mathcal{A}) = \mathcal{A}$.
2. Consider $\Omega = \{a, b, c\}$ and $M = \{\emptyset, \{a\}, \{b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$. The σ-field generated by M is $\sigma(M) = \mathcal{P}(\Omega)$.
3. Let (X, d) be a metric space and denote by $S \coloneqq \{A \subseteq X : A \text{ open}\}$ the set of all open subsets of X. The Borel sigma field, $B(X)$, is a sigma field generated by S, i.e., $B(X) \coloneqq \sigma(S)$.

Definition B.21. Let Ω be a set and $\mathcal{A}$ a σ-field on Ω. The pair $(\Omega, \mathcal{A})$ is called a measurable space. Elements of $\mathcal{A}$ are called measurable sets.

Examples of measurable sets include all pairs of $(\Omega, \mathcal{A})$ in the previous examples, where $\mathcal{A}$ was a σ-field on Ω.

Definition B.22. A measure on a measurable space $(\Omega, \mathcal{A})$ is a function $\mu : \mathcal{A} \to \mathbb{R}_{\geq 0} \cup +\infty$ with

- $\mu(\emptyset) = 0$

- If $(A_i)_{i\geq 1}$ is a countable sequence of pairwise disjoint elements of $\mathcal{A}$, then

$$\mu\left(\bigcup_{i\geq 1} A_i\right) = \sum_{i\geq 1} \mu(A_i)$$

Examples.

1. The counting measure on $(\Omega, \mathcal{A})$ is defined for $A \in \mathcal{A}$ by

$$\mu(A) = \begin{cases} |A| & : A \text{ is finite} \\ +\infty & : \text{else} \end{cases}$$

   where $|A|$ denotes the cardinality of A.

2. Let $(\Omega, \mathcal{A})$ be a measurable space and $x \in \Omega$. The Dirac measure at x is defined for $A \in \mathcal{A}$ by

$$\delta_x(A) = \begin{cases} 1 & : x \in A \\ 0 & : \text{else} \end{cases}$$

3. The Lebesgue measure λ on the measurable space $(\mathbb{R}, B(\mathbb{R}))$ is the unique measure satisfying $\lambda([a,b]) = b - a$ for all $a \leq b$.

Definition B.23. Let $(\Omega, \mathcal{A})$ be a measurable space, and μ a measure on $(\Omega, \mathcal{A})$. Then $(\Omega, \mathcal{A}, \mu)$ is called a measure space.

Definition B.24. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $\mu(\Omega) = 1$. Then μ is a probability measure, and $(\Omega, \mathcal{A}, \mu)$ a probability space.

Probability measures are often denoted by $\mathbb{P}$.

Definition B.25. Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be two measurable spaces. A function $f : X \to Y$ is called a $\mathcal{A}_X$-$\mathcal{A}_Y$-measurable or simply measurable if for all $A \in \mathcal{A}_Y$ it holds that

$$f^{-1}(A) \coloneqq \{x \in X : f(x) \in A\} \in \mathcal{A}_X$$

Examples.

1. Consider $\Omega = \{a, b, c\}$ with the σ-field $\mathcal{A} = \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\}$. Define $f : (\Omega, \mathcal{A}) \to (\Omega, \mathcal{A})$ by $f(a) = a$, $f(b) = c$, $f(c) = b$. Then $f^{-1}(\{a\}) = \{a\} \in \mathcal{A}$, $f^{-1}(\{b, c\}) = \{b, c\} \in \mathcal{A}$, and $f^{-1}(\{a, b, c\}) = \{a, b, c\} \in \mathcal{A}$ so f is measurable.

2. Now define $f : (\Omega, \mathcal{A}) \to (\Omega, \mathcal{A})$ by $f(a) = b$, $f(b) = a$, $f(c) = c$. Then $f^{-1}(\{a\}) = \{b\} \notin \mathcal{A}$, so f is not measurable.

3. Let $(\Omega, \mathcal{A})$ be a measurable space, equip $\{0, 1\}$ with the σ-field $\mathcal{P}(\{0, 1\})$, and

let $A \in \mathcal{A}$. The indicator function $\mathbf{1}_A : \Omega \to \{0,1\}$ defined by

$$\mathbf{1}_A(\omega) \coloneqq \begin{cases} 1 & : \omega \in A \\ 0 & : \text{else} \end{cases}$$

is a $\mathcal{A}$-$\mathcal{P}(\{0,1\})$-measurable function.

Definition B.26. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A random variable with values in E is a measurable function $X : \Omega \to E$.

Definition B.27. Let X be a random variable with values in $(E, \mathcal{E})$. The law $\mathbb{P}_X$ of the random variable X is a probability measure on $(E, \mathcal{E})$ defined by

$$\mathbb{P}_X(B) \coloneqq \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\})$$

for $B \in \mathcal{E}$. We also call $\mathbb{P}_X$ the probability distribution of X.

To build up to L^p spaces, we first focus on introducing the Lebesgue integral. Before we can integrate general measurable functions, we first define the integral for a family of simple functions.

Definition B.28. Let $(\Omega, \mathcal{A})$ be a measurable space. The function $\varphi : \Omega \to \mathbb{R}$ is called a simple function if there exists pairwise distinct $a_1, \ldots, a_k \in \mathbb{R}$ and pairwise disjoint $A_1, \ldots, A_k \in \mathcal{A}$, with $\bigcup_{i=1}^k A_i = \Omega$, such that for all $\omega \in \Omega$ the function φ can be written as

$$\varphi(\omega) = \sum_{i=1}^{k} a_i \mathbf{1}_{A_i}(\omega)$$

Examples.

1. On $(\mathbb{R}, B(\mathbb{R}))$ the function

   $$\varphi = 0 \cdot \mathbf{1}_{(-\infty,0)\cup(4,\infty)} + 2 \cdot \mathbf{1}_{[0,1]} + 3 \cdot \mathbf{1}_{(1,4]}$$

   is a simple function.

2. On $(\mathbb{R}, B(\mathbb{R}))$ the function $f(x) = x$ is not simple, as it takes up infinitely many values.

Remark B.3. Let $(\Omega, \mathcal{A})$ be a measurable space, and $f : \Omega \to [0, \infty]$ be a measurable function. It can be shown that there exists a sequence $(\varphi_i)_{i\in\mathbb{N}}$ of non-negative simple functions such that

1. For all $\omega \in \Omega$ it holds that $0 \leq \varphi_1(\omega) \leq \varphi_2(\omega) \leq \ldots$
2. The sequence $(\varphi_i(\omega))_{i\in\mathbb{N}}$ converges to $f(\omega)$ for all $\omega \in \Omega$.

Definition B.29. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ a non-negative simple function. The Lebesgue integral of φ with respect to μ is defined as

$$\int_{\Omega} \varphi d\mu \coloneqq \sum_{i=1}^{n} a_i \mu(A_i)$$

Remark B.4. By convention, we use $\infty \cdot 0 = 0$.

Examples.

- On $(\mathbb{R}, B(\mathbb{R}), \lambda)$ the function

  $$\varphi = 0 \cdot \mathbf{1}_{(-\infty,0)\cup(4,\infty)} + 2 \cdot \mathbf{1}_{[0,1]} + 3 \cdot \mathbf{1}_{(1,4]}$$

  has a Lebesgue integral

  $$\int_{\mathbb{R}} \varphi d\lambda = 0\lambda(\mathbb{R} \setminus [0,4]) + 2\lambda([0,1]) + 3\lambda((1,4]) = 11$$

- Let Ω be a set and $\omega \in \Omega$. Furthermore consider the measure space $(\Omega, \mathcal{A}, \delta_\omega)$ and let $\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ be a non-negative simple function. Denote by j the index for which $\omega \in A_j$. Then

  $$\int_{\Omega} \varphi d\delta_\omega = \sum_{i} a_i \delta_\omega(A_i) = a_j = \varphi(\omega)$$

Definition B.30. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $f : \Omega \to [0, \infty]$ a non-negative measurable function. The Lebesgue integral of f with respect to μ is defined as

$$\int_{\Omega} f d\mu = \sup \left\{ \int_{\Omega} \varphi d\mu : \varphi \text{ simple function}, 0 \leq \varphi \leq f \right\}$$

Remark B.5. Due to Remark B.3, it can be shown that this supremum is attained as a limit.

Definition B.31. Let $(\Omega, \mathcal{A})$ be a measurable space, and $f : \Omega \to \mathbb{R}$ be a measurable function. The positive and negative parts of f are defined respectively as

$$f^+(\omega) \coloneqq \max(f(\omega), 0) \qquad f^-(\omega) \coloneqq \max(-f(\omega), 0)$$

Both f^+ and f^- are non-negative measurable functions, and $f = f^+ - f^-$.

Definition B.32. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $f : \Omega \to \mathbb{R}$ a measurable function. If at least one of $\int_\Omega f^+ d\mu$ or $\int_\Omega f^- d\mu$ is finite, we define the Lebesgue integral of f by

$$\int_{\Omega} f d\mu = \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu$$

Definition B.33. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. A property P is said to hold almost everywhere (a.e.) with respect to μ if there exists a set $S \in \mathcal{A}$ with $\mu(S) = 0$ such that P holds for all $\omega \in \Omega \setminus S$.

Definition B.34. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. For $p \in \mathbb{N}_+$ the space L^p is defined as

$$L^p(\Omega, \mathcal{A}, \mu) \coloneqq \left\{ f : \Omega \to \mathbb{R} : f \text{ is measurable and } \int_\Omega |f|^p d\mu < \infty \right\}$$

where two elements f, g are considered equivalent if $f = g$ almost everywhere.

Definition B.35. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $f : \Omega \to \mathbb{R}$. The L^p norm is defined by

$$\|f\|_p = \left(\int_\Omega |f|^p d\mu \right)^{\frac{1}{p}}$$

Remark B.6. It can be shown that the pair $(L^p, \|\cdot\|_p)$ forms a Banach space. Furthermore, if we define the inner product $\langle \cdot, \cdot \rangle_{L^2}$ on $L^2(\Omega, \mathcal{A}, \mu)$ as

$$\langle f, g \rangle_{L^2} = \int_\Omega (f \cdot g) d\mu,$$

the pair $(L^2(\Omega, \mathcal{A}, \mu), \langle \cdot, \cdot \rangle_{L^2})$ forms a Hilbert space.

We can generalize L^p spaces to vector valued functions in the following way.

Definition B.36. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $\|\cdot\|_{\mathbb{R}^m}$ a norm on $\mathbb{R}^m$. For $p \in \mathbb{N}_+$ the space L^p of $\mathbb{R}^m$ valued functions is defined as

$$L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m) \coloneqq \left\{ f : \Omega \to \mathbb{R}^m : f \text{ is measurable and } \int_\Omega \|f\|_{\mathbb{R}^m}^p \, d\mu < \infty \right\}$$

where two elements f, g are considered equivalent if $f = g$ almost everywhere.

Remark B.7. Let X be a normed space equipped with two norms $\|\cdot\|, \|\cdot\|'$. The norms are said to be equivalent if there exists $C_1, C_2 > 0$ such that for all $x \in X$ it holds that

$$C_1 \|x\|' \leq \|x\| \leq C_2 \|x\|'.$$

It can be shown that on $\mathbb{R}^m$ all norms are equivalent, and therefore the choice of norm on $\mathbb{R}^m$ does not affect $L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m)$.

Definition B.37. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space, $f : \Omega \to \mathbb{R}^m$ measurable, and $\|\cdot\|_{\mathbb{R}^m}$ a norm on $\mathbb{R}^m$. For $p \in \mathbb{N}_+$, the L^p norm is defined by

$$\|f\|_p \coloneqq \left(\int_\Omega \|f(\omega)\|_{\mathbb{R}^m}^p \, d\mu(\omega) \right)^{1/p}.$$

Remark B.8. It can be shown that the pair $(L^p(\Omega, \mathcal{A}, \mu; \mathbb{R}^m), \|\cdot\|_p)$ forms a Banach space. Furthermore, if we equip $\mathbb{R}^m$ with the Euclidean inner product $\langle y, z\rangle_{\mathbb{R}^m} = \sum_{i=1}^m y_i z_i$ and define the inner product $\langle \cdot, \cdot \rangle_{L^2}$ on $L^2(\Omega, \mathcal{A}, \mu; \mathbb{R}^m)$ by

$$\langle f, g\rangle_{L^2} \coloneqq \int_\Omega \langle f(\omega), g(\omega)\rangle_{\mathbb{R}^m} \, d\mu(\omega),$$

then $(L^2(\Omega, \mathcal{A}, \mu; \mathbb{R}^m), \langle \cdot, \cdot \rangle_{L^2})$ forms a Hilbert space.